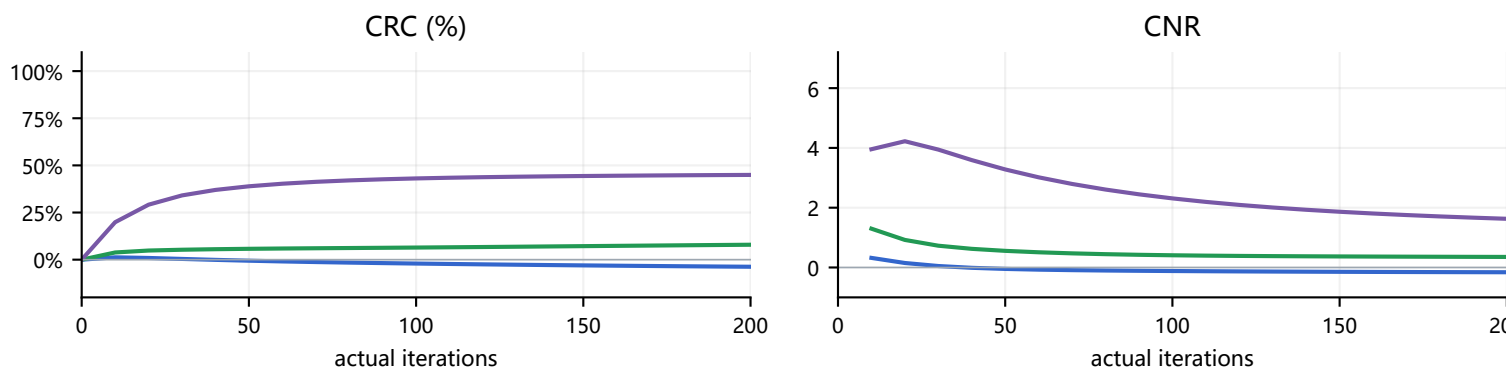
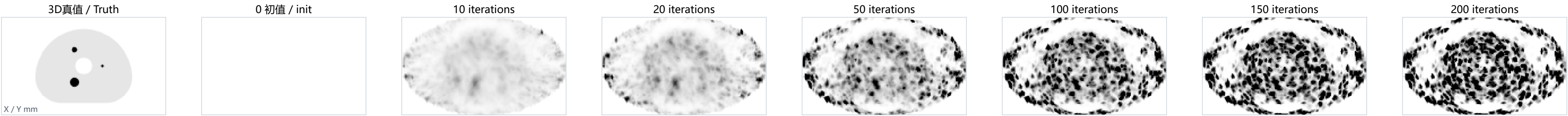


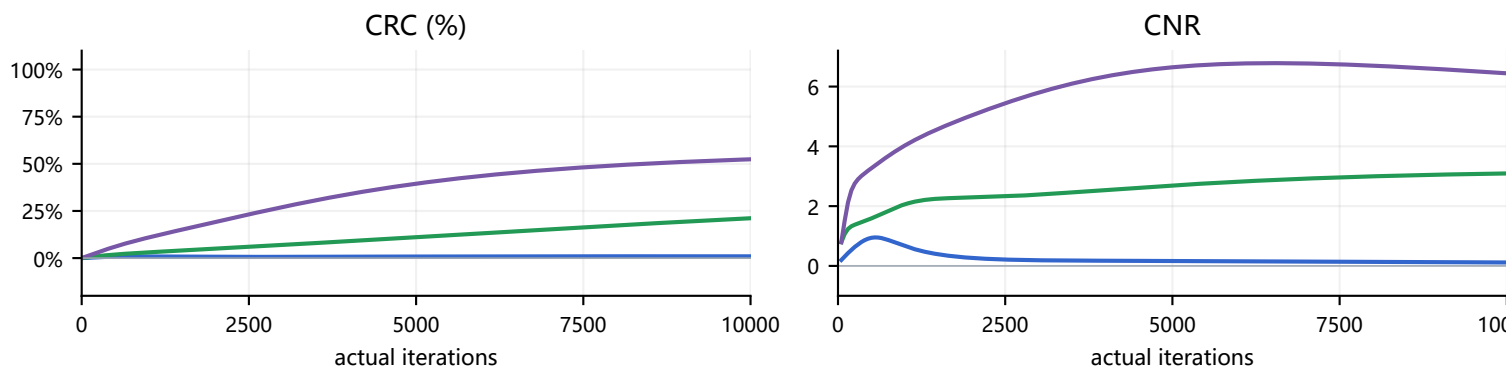
整体迭代对比 / 72 mm axial MIP

EHE 0-200 | JSCC 0-10000 | 9 result routes | independent iteration ranges; columns do not imply equal convergence

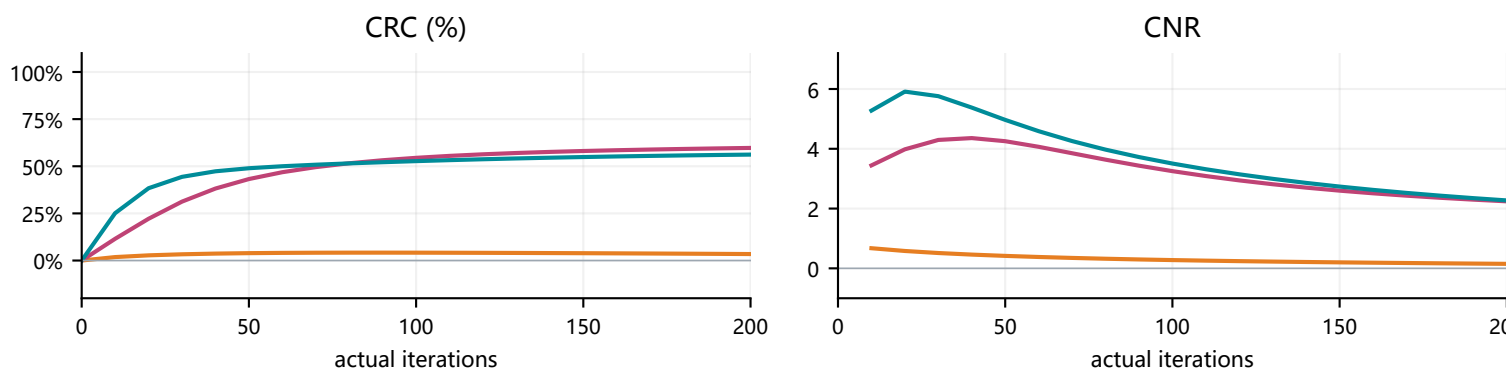
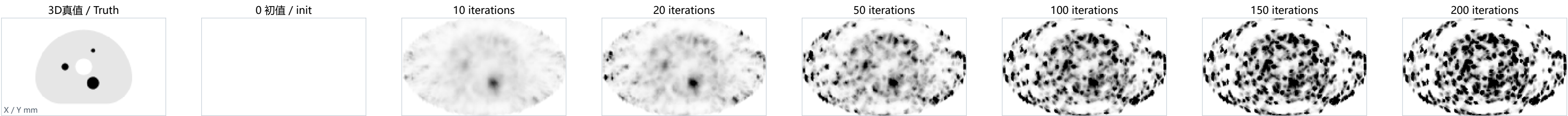
EHE
218校正单光子
218 corrected single
0-200



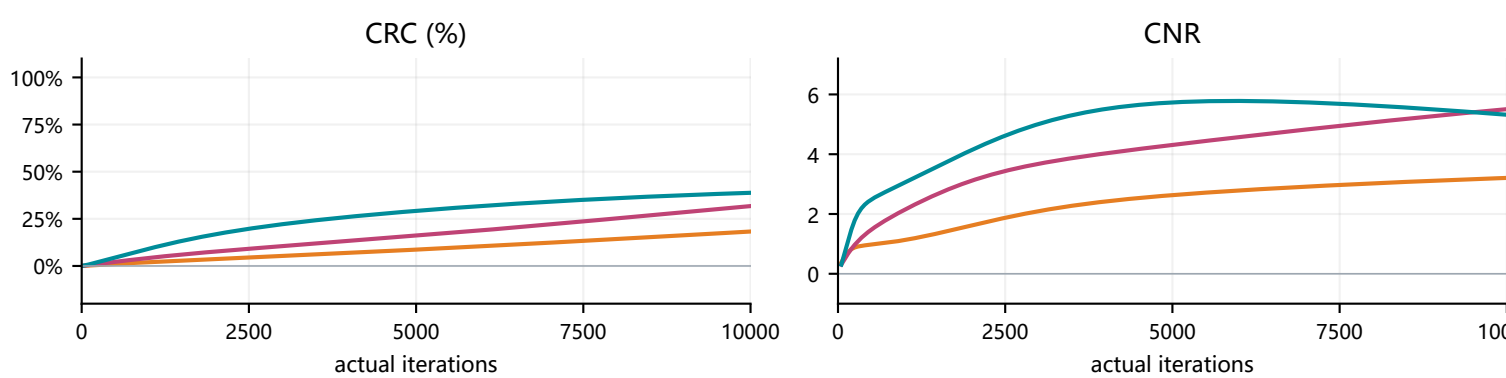
JSCC
218校正单光子
218 corrected single
0-10000



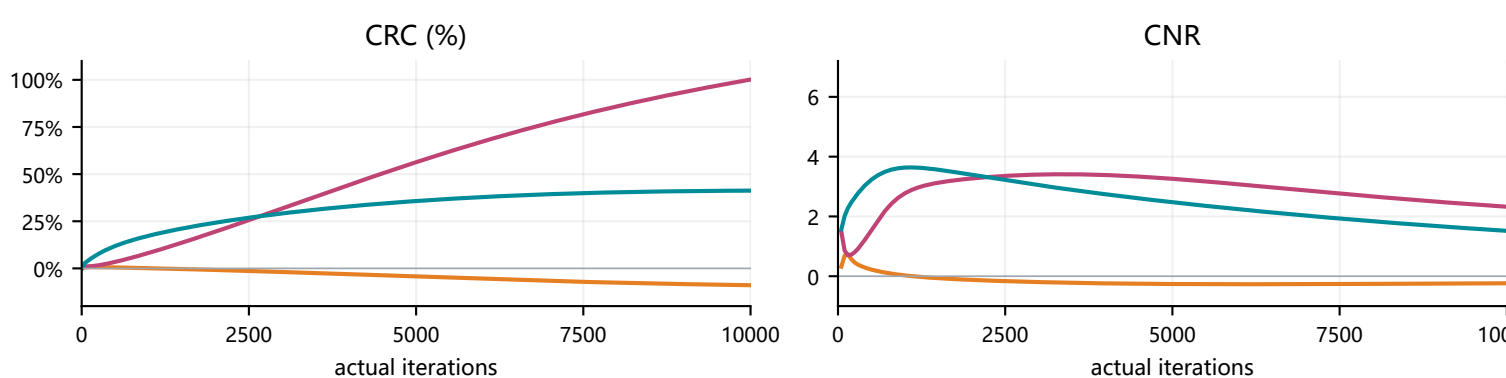
EHE
440单光子
440 single
0-200



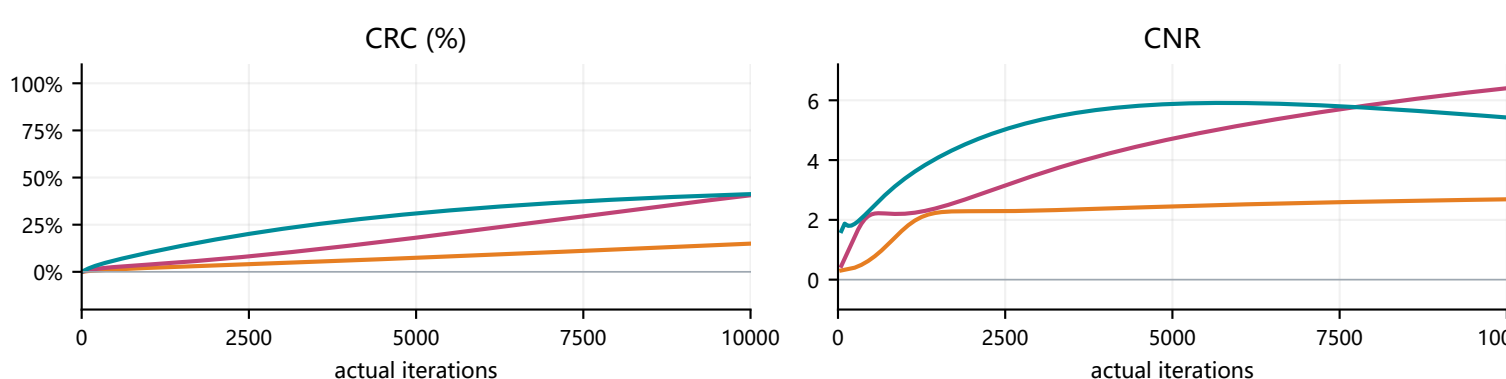
JSCC
440单光子
440 single
0-10000



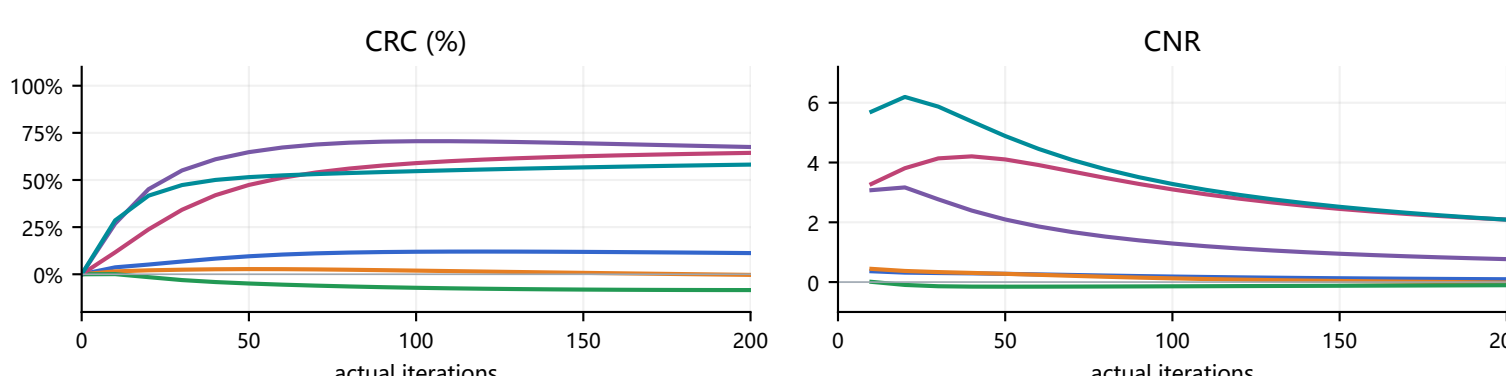
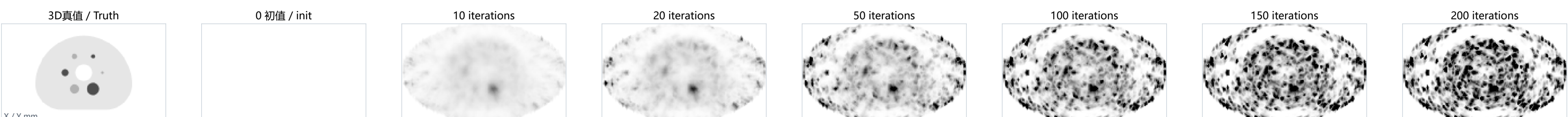
JSCC
440 SC Compton
440 SC Compton
0-10000



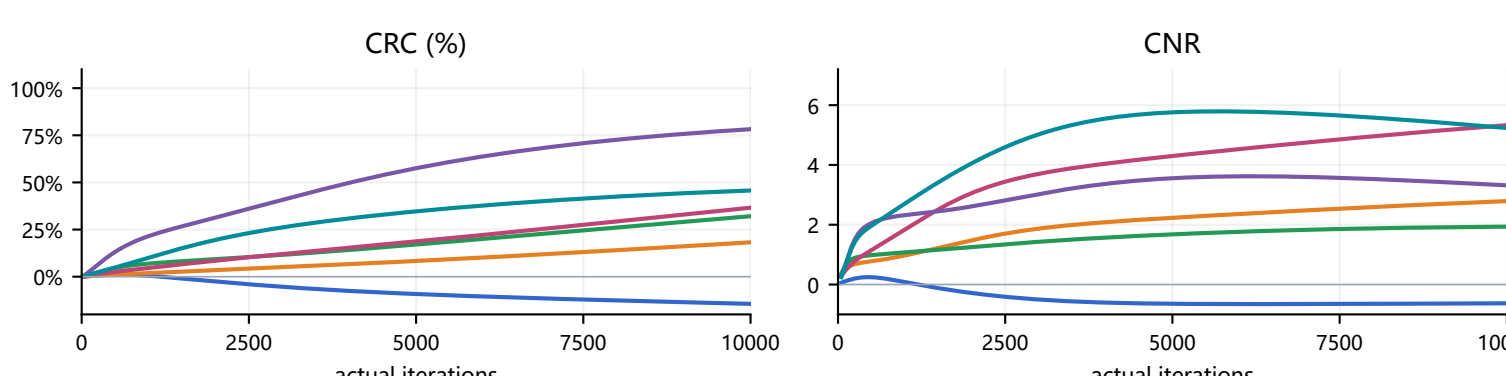
JSCC
440 JSCC
440 JSCC
0-10000



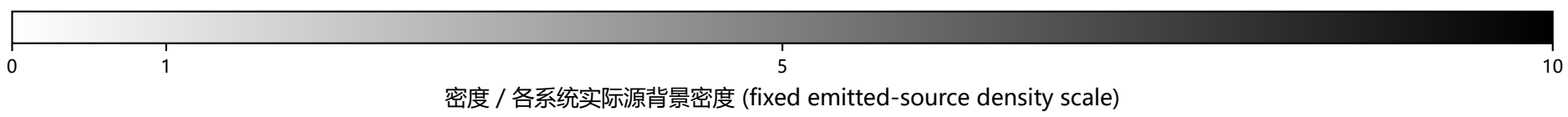
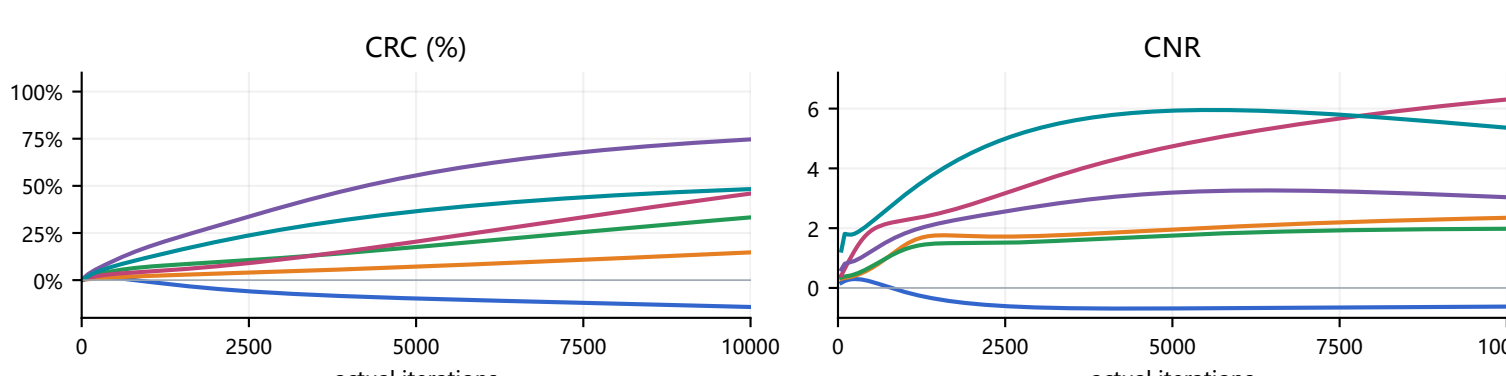
EHE
双能单光子和
dual single sum
0-200



JSCC
双能单光子和
dual single sum
0-10000



JSCC
440 JSCC + 218单光子和
440 JSCC + 218 single
0-10000



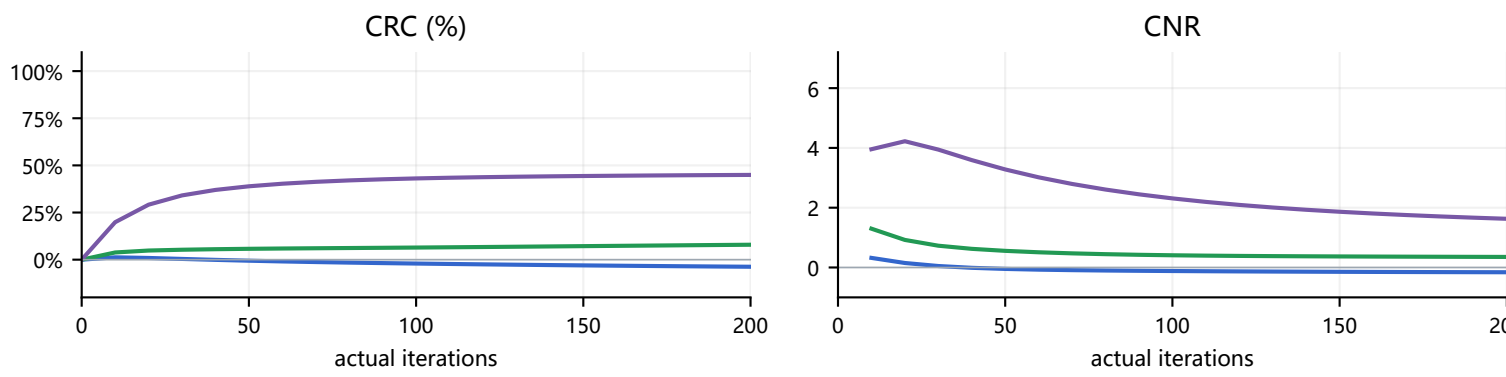
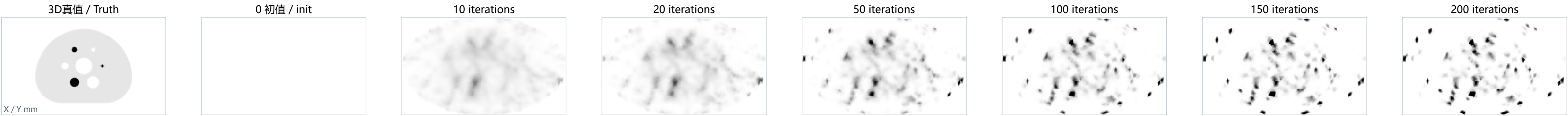
0 = known all-ones initialization (single 1, sum 2 gamma/mm³), not a saved frame; CNR at 0 undefined.
Crop 0; no smoothing or fitted gain; fixed gray_r 0-10. Metrics: native 120 mm / real 3D ROI. EHE physical audit remains HOLD.
218 background budget: EHE440 final200; JSCC440 final10000. Detector/material/count differences are not solely algorithm effects.

真实3D球ROI / Sphere diameter
10 mm 13 mm 17 mm 22 mm 28 mm 37 mm

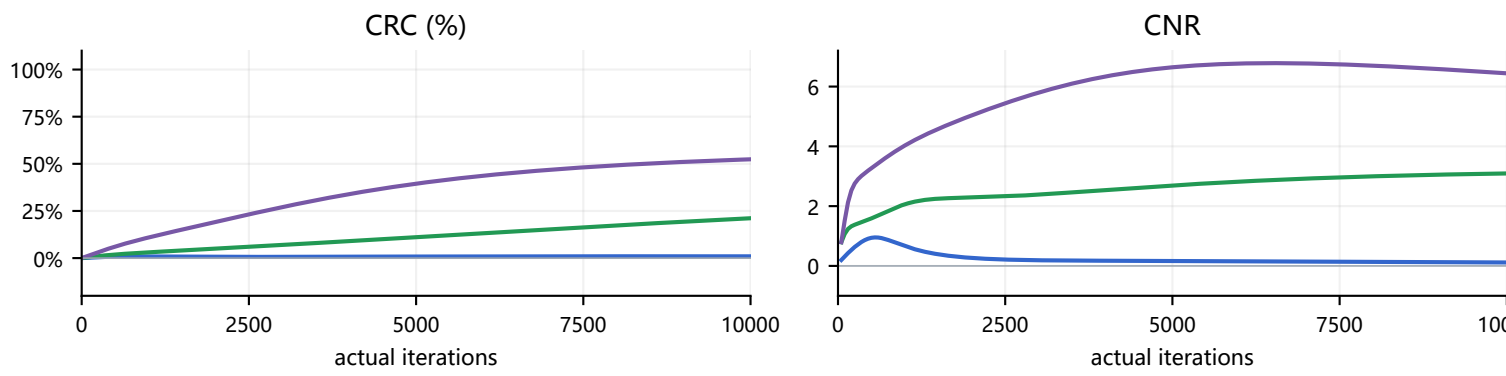
整体迭代对比 / Axial z=1.5 mm

EHE 0-200 | JSCC 0-10000 | 9 result routes | independent iteration ranges; columns do not imply equal convergence

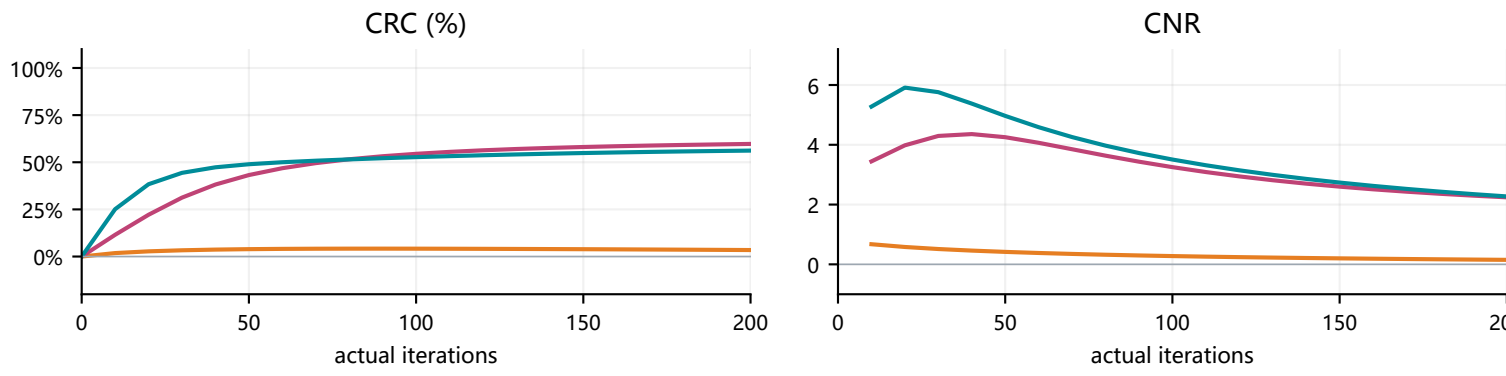
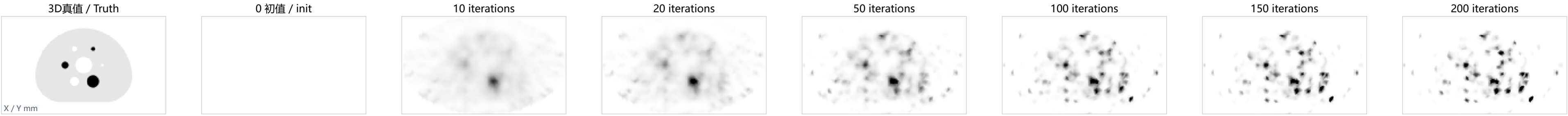
EHE
218校正单光子
218 corrected single
0-200



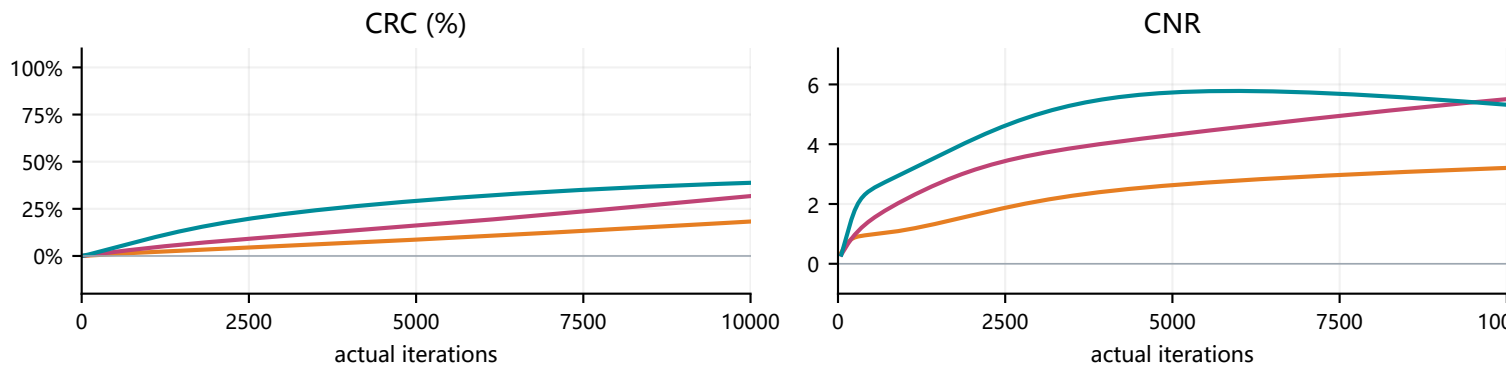
JSCC
218校正单光子
218 corrected single
0-10000



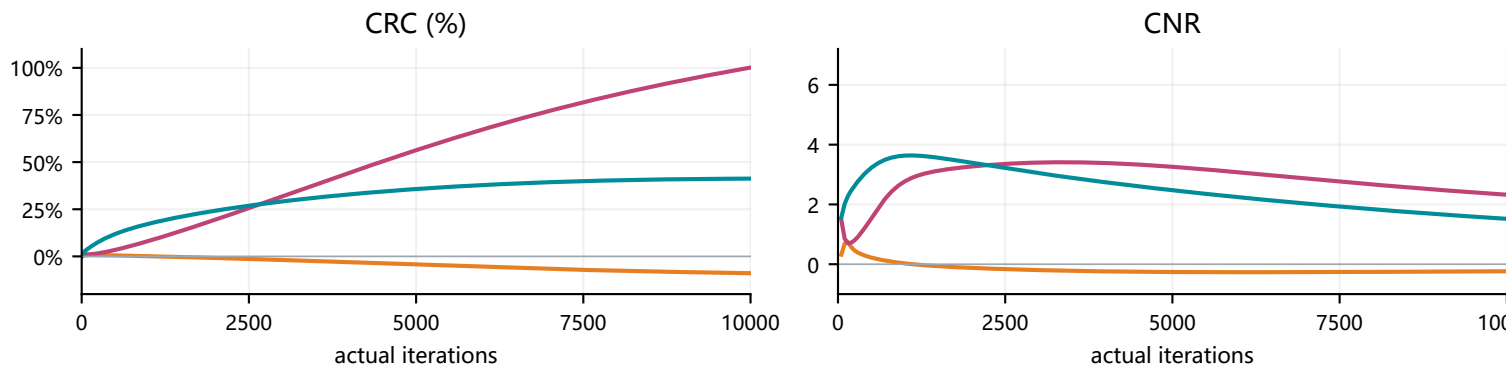
EHE
440单光子
440 single
0-200



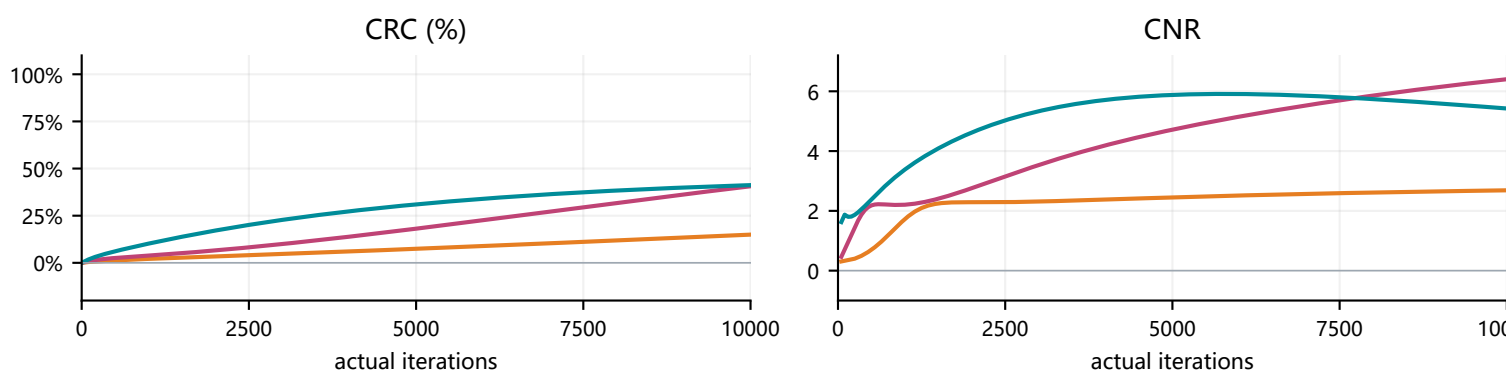
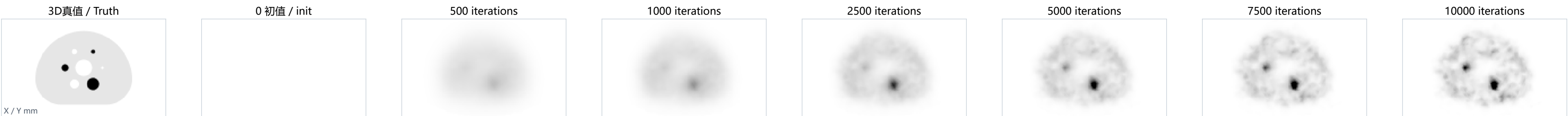
JSCC
440单光子
440 single
0-10000



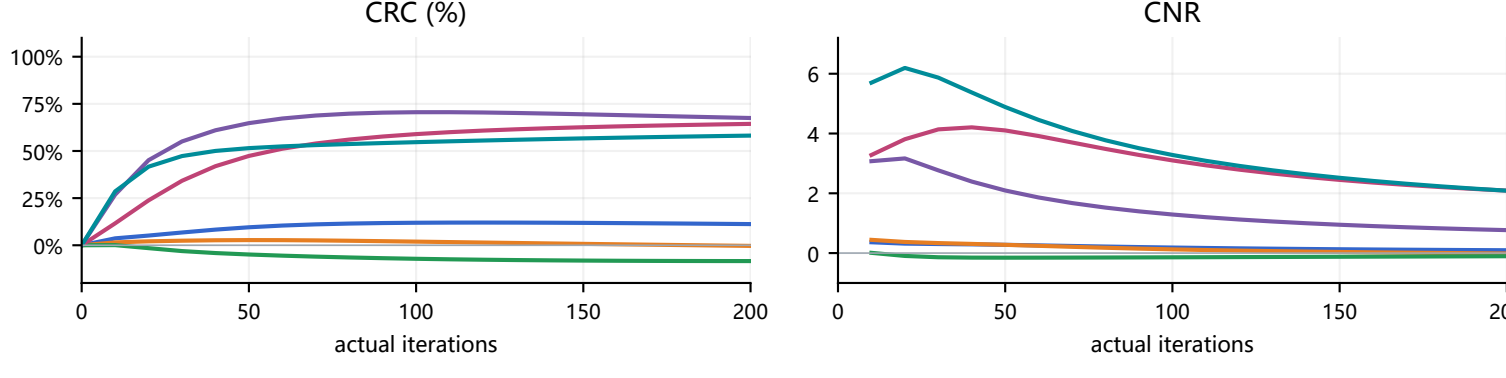
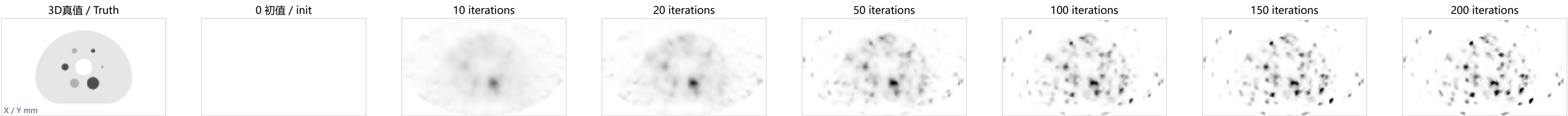
JSCC
440 SC Compton
440 SC Compton
0-10000



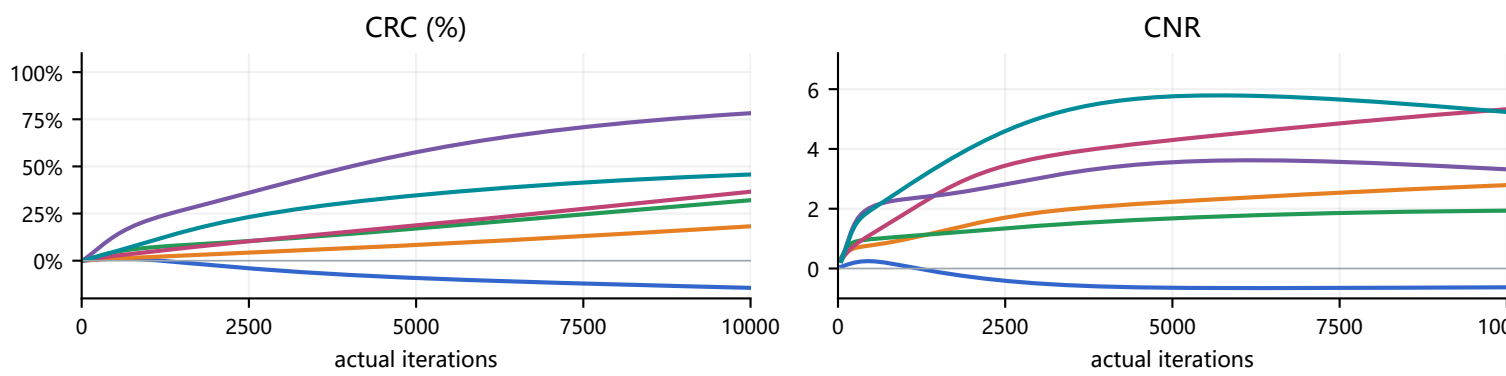
JSCC
440 JSCC
440 JSCC
0-10000



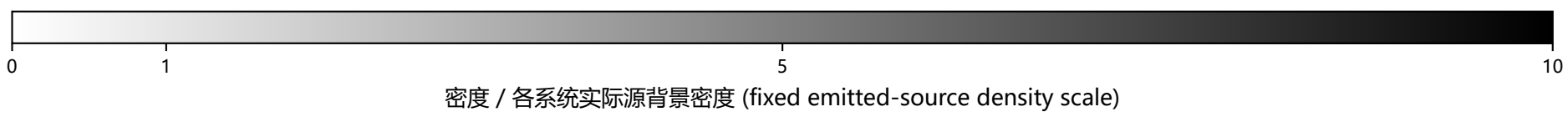
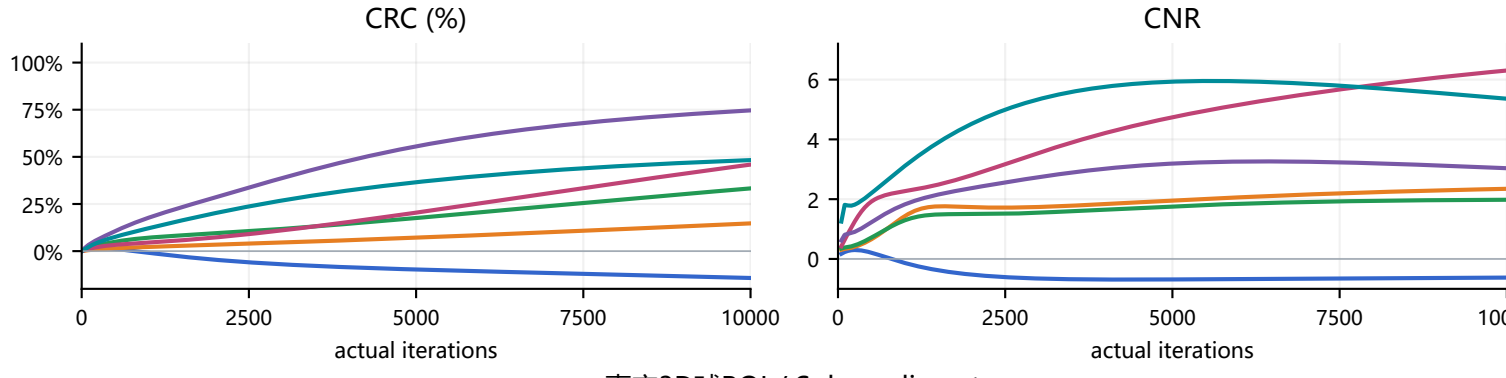
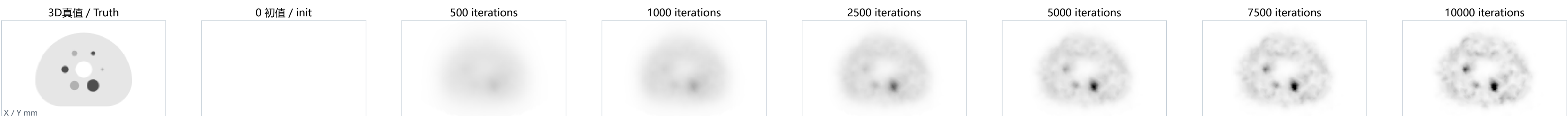
EHE
双能单光子和
dual single sum
0-200



JSCC
双能单光子和
dual single sum
0-10000



JSCC
440 JSCC + 218单光子和
440 JSCC + 218 single
0-10000

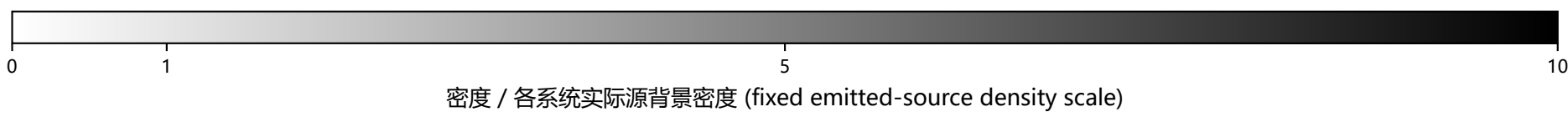
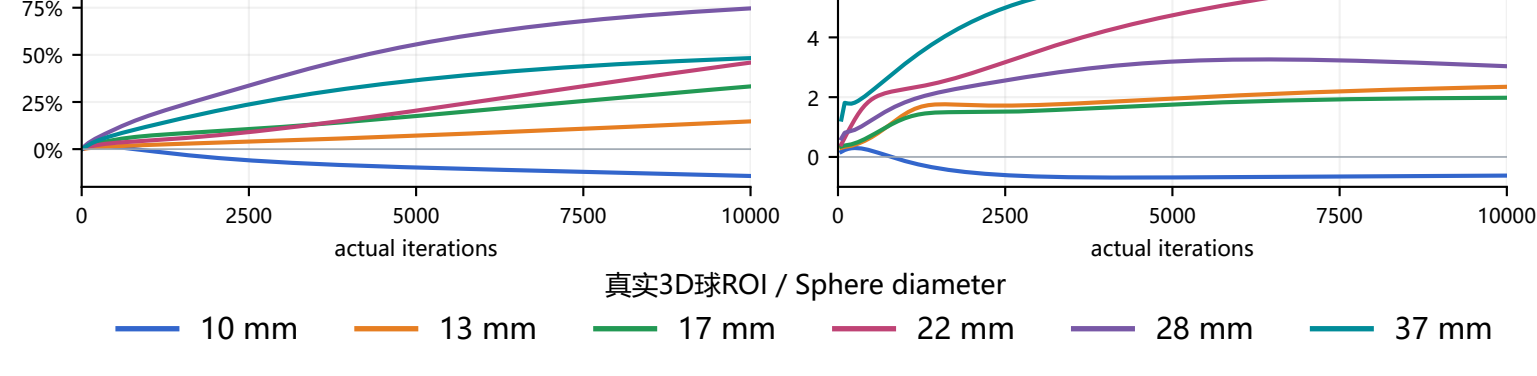
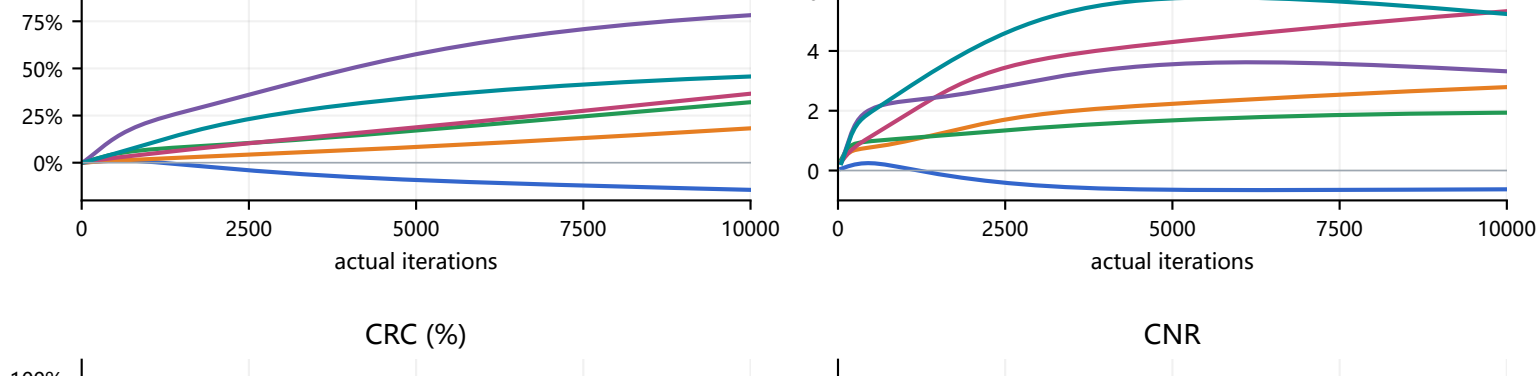
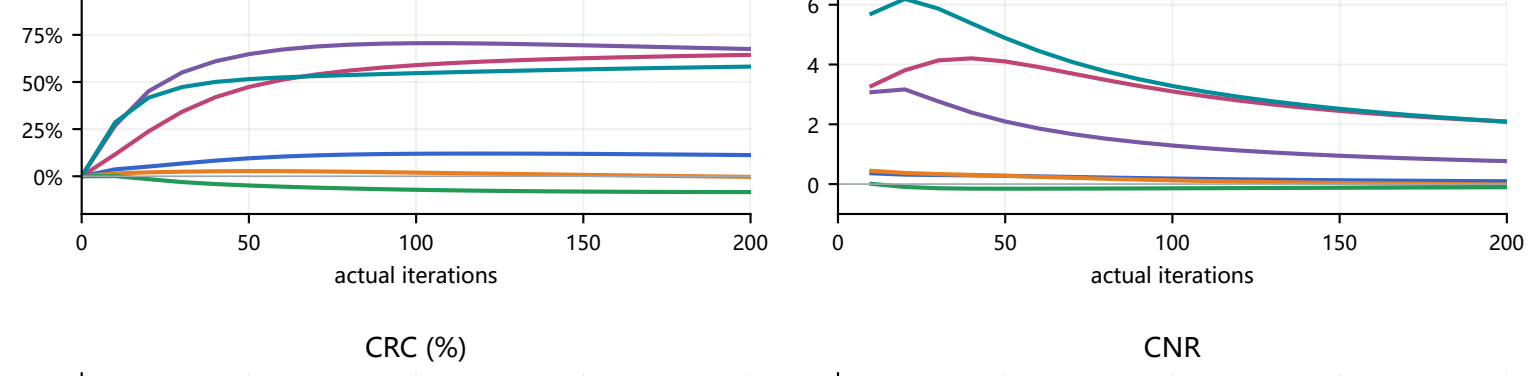
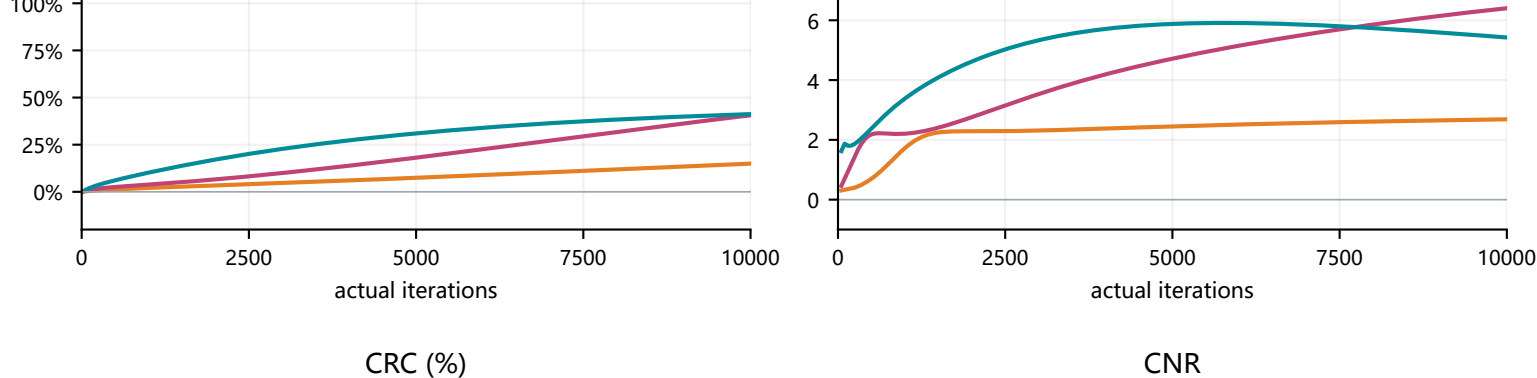
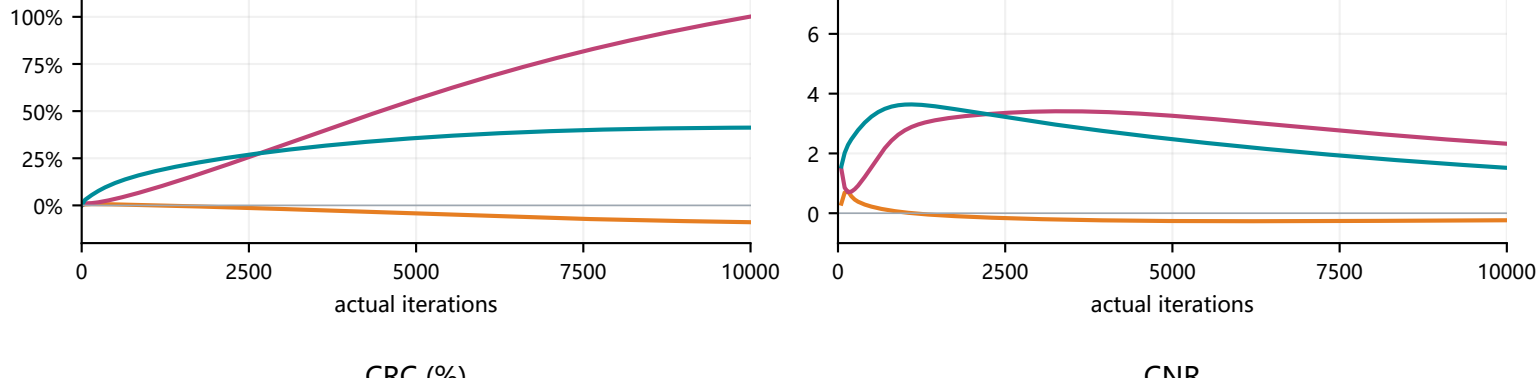
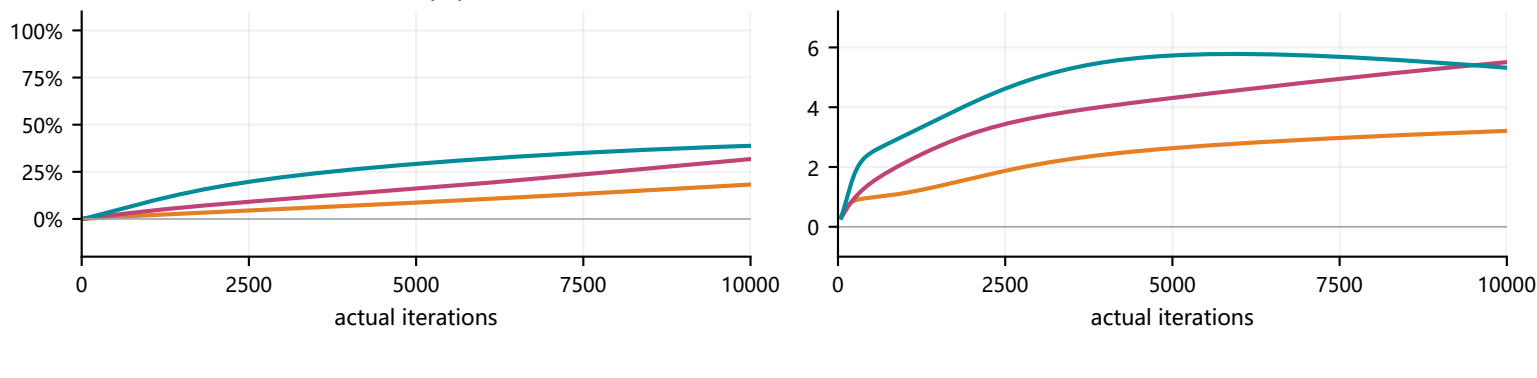
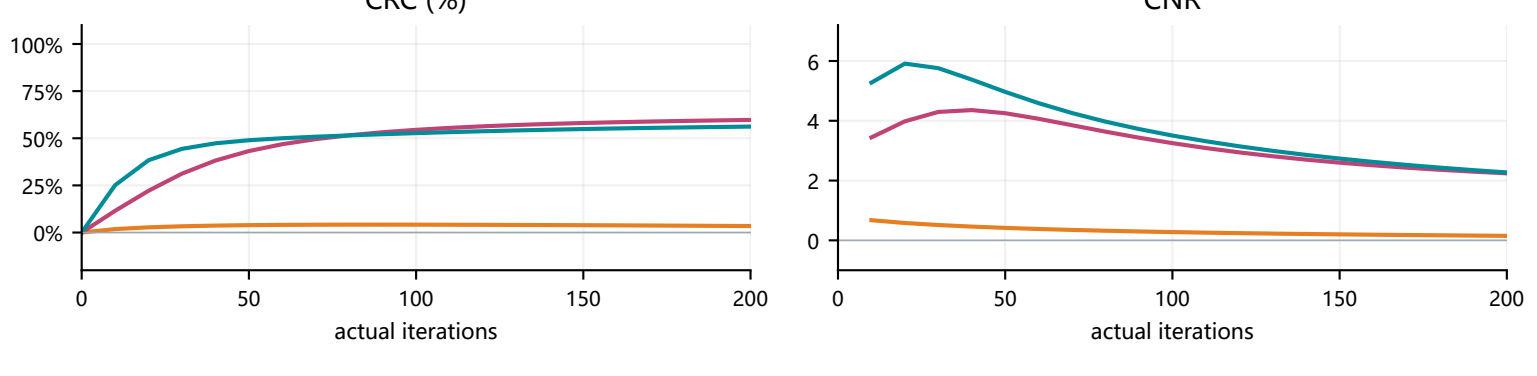
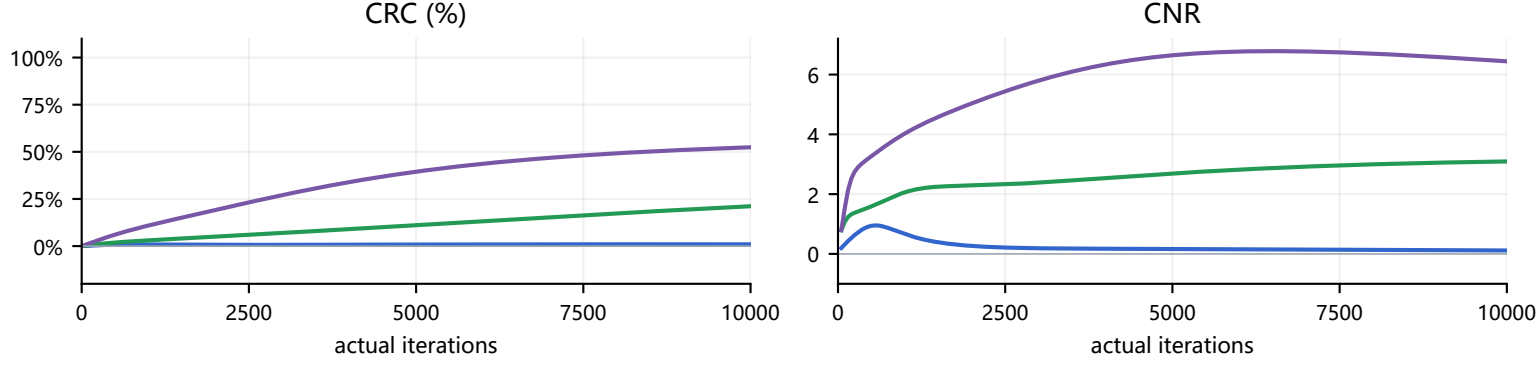
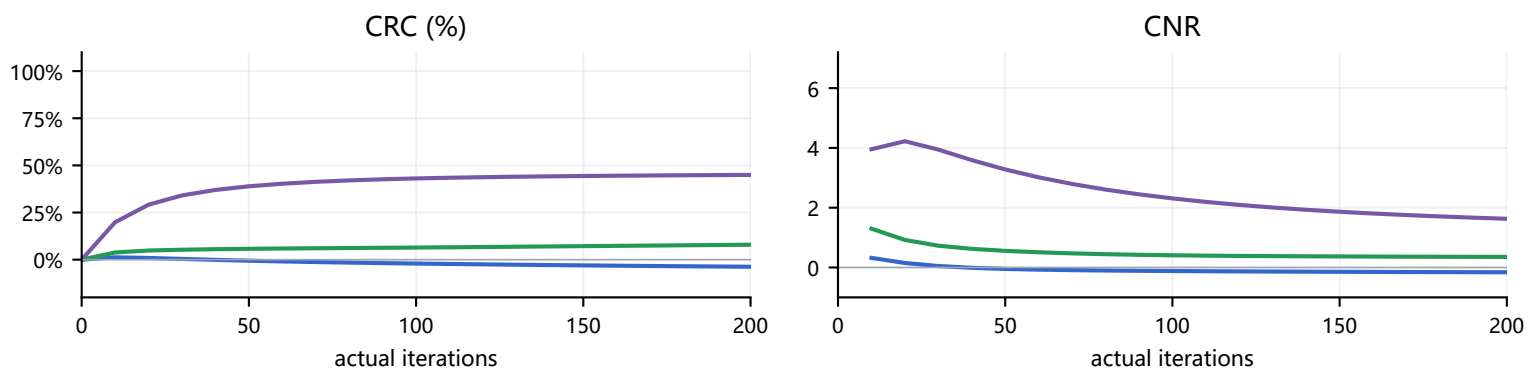
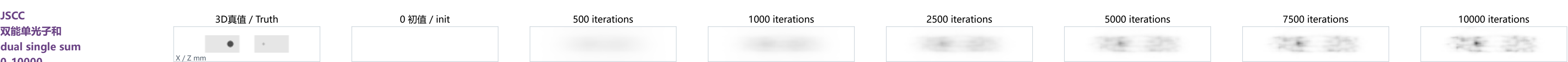
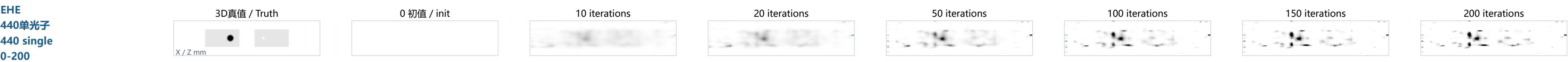
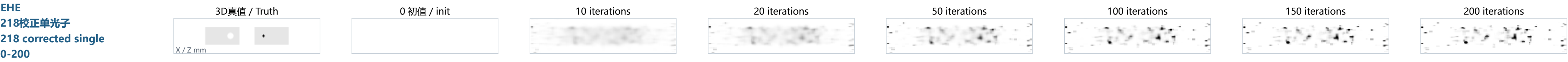


0 = known all-ones initialization (single 1, sum 2 gamma/mm³), not a saved frame; CNR at 0 undefined.
Crop 0; no smoothing or fitted gain; fixed gray_r 0-10. Metrics: native 120 mm / real 3D ROI. EHE physical audit remains HOLD.
218 background budget: EHE440 final200; JSCC440 final10000. Detector/material/count differences are not solely algorithm effects.

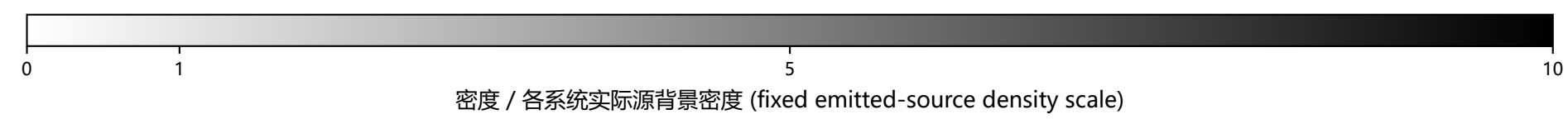
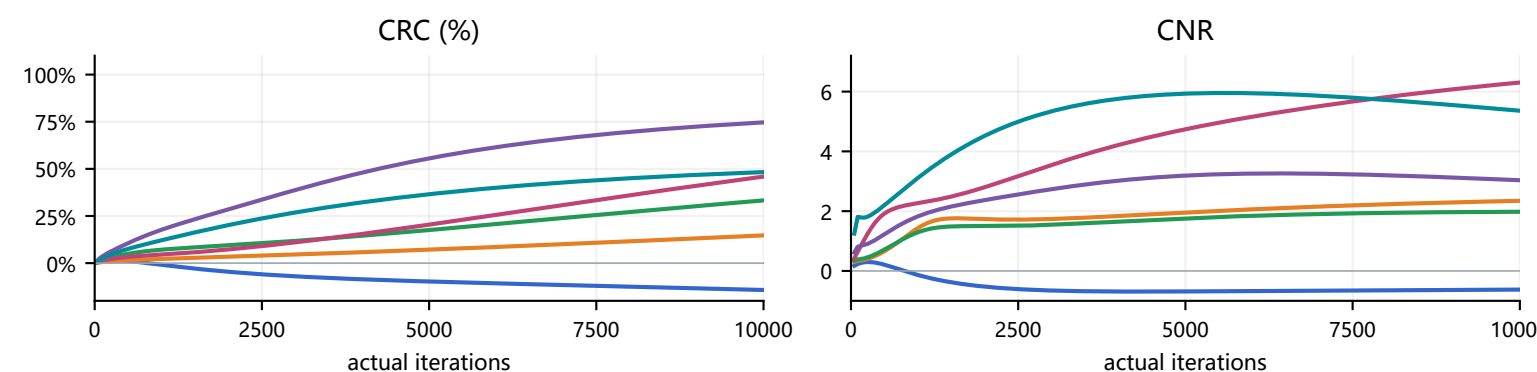
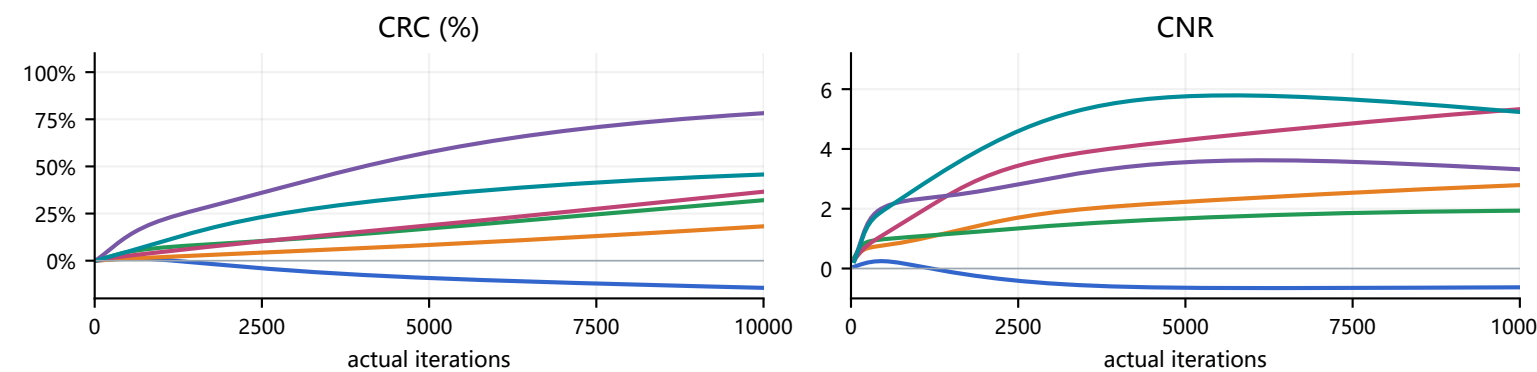
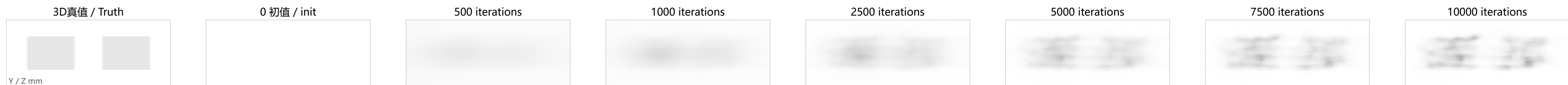
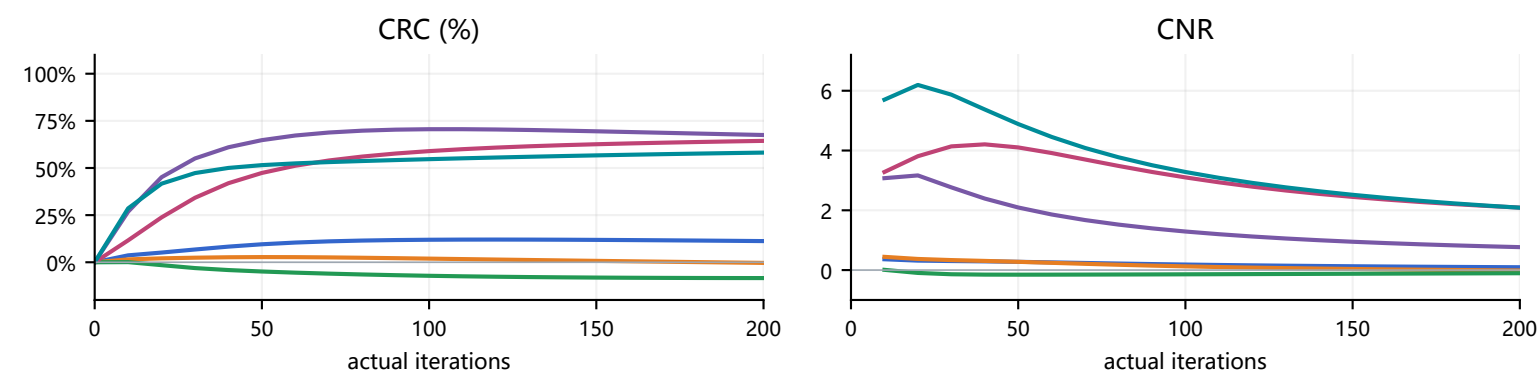
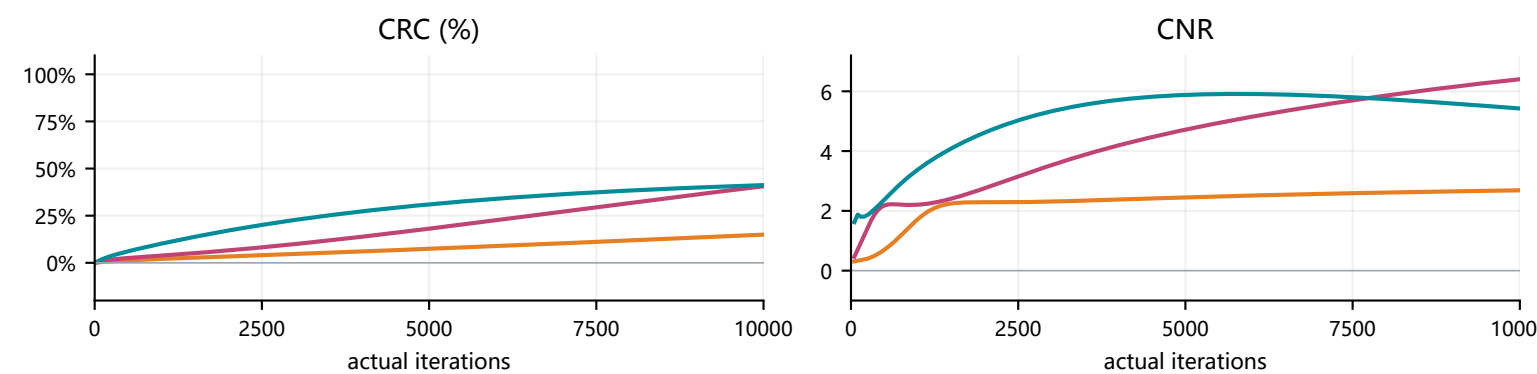
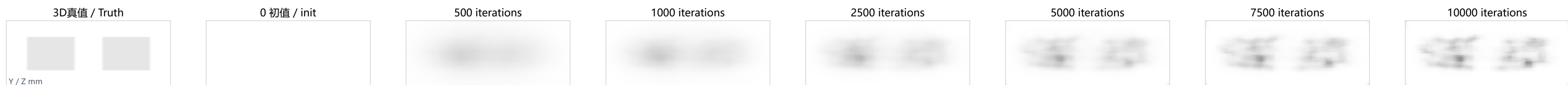
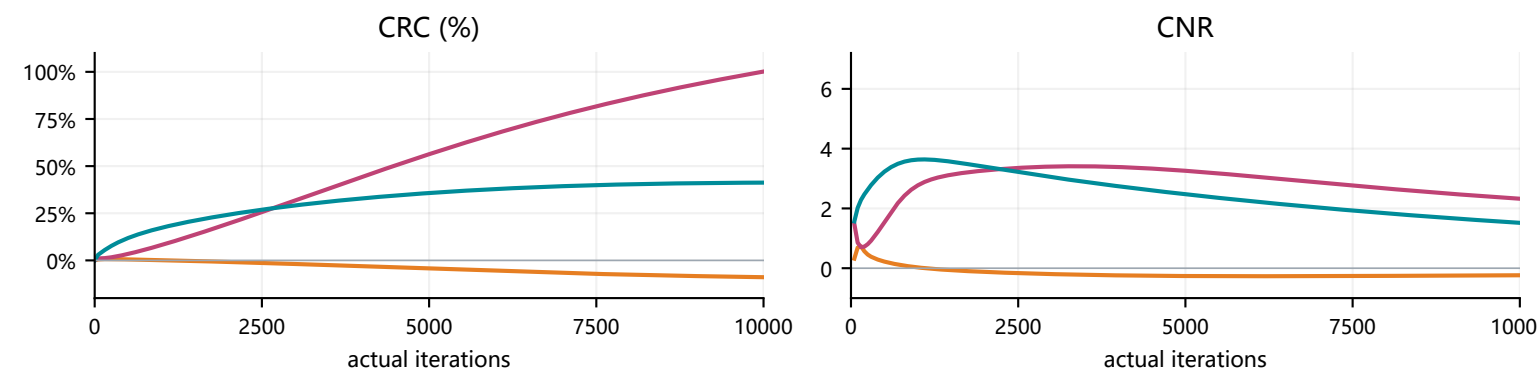
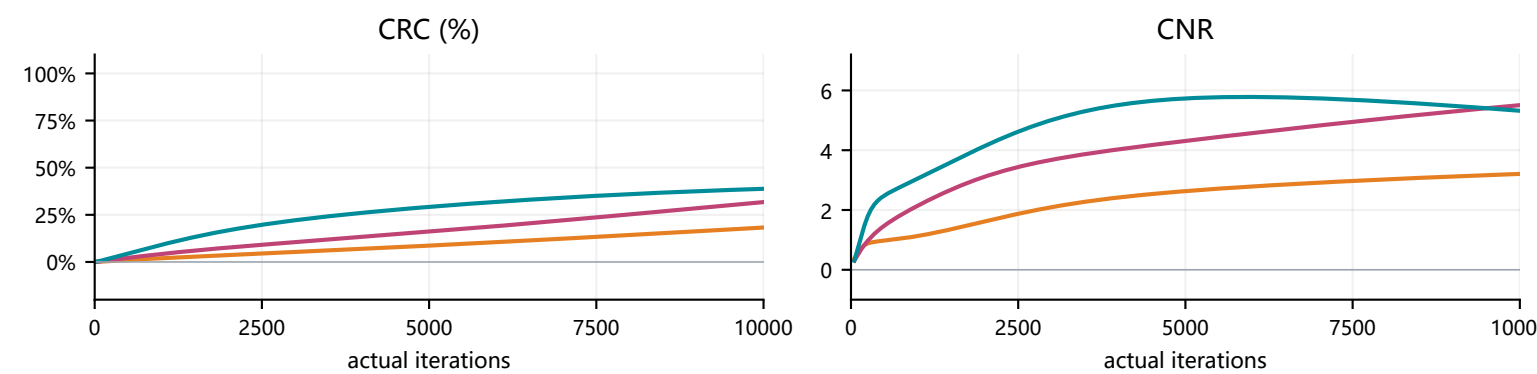
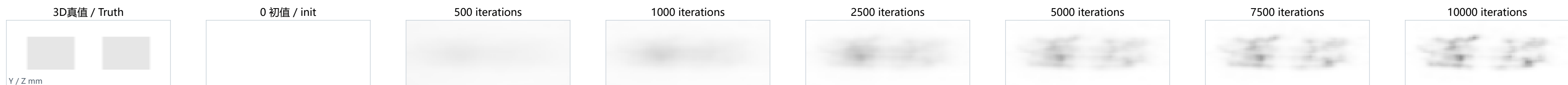
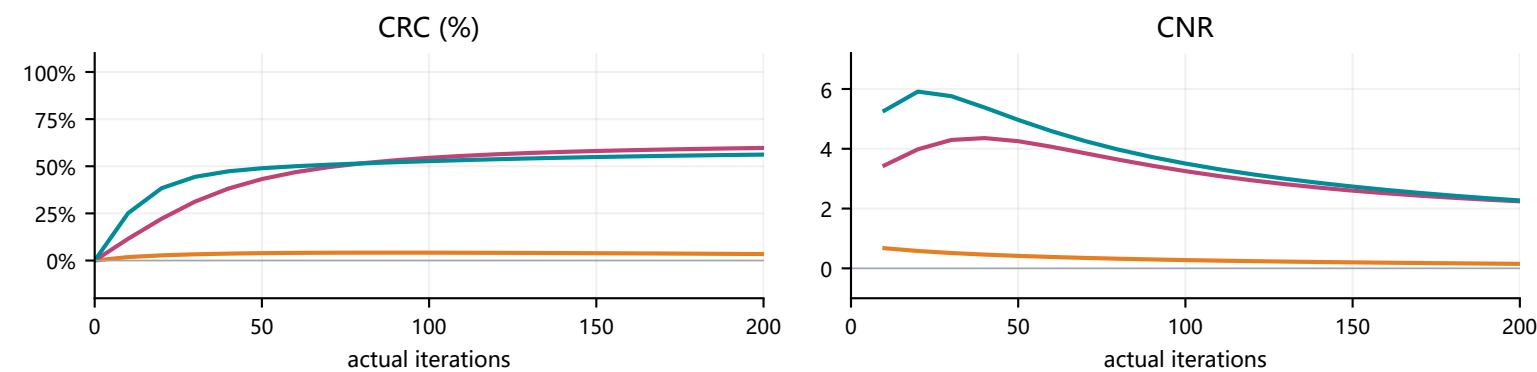
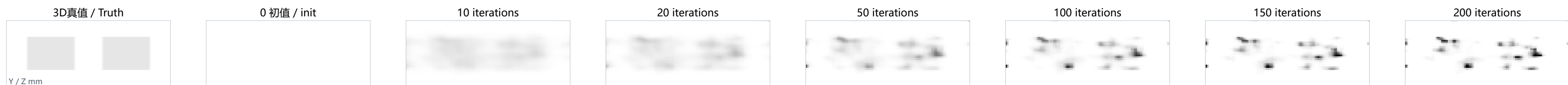
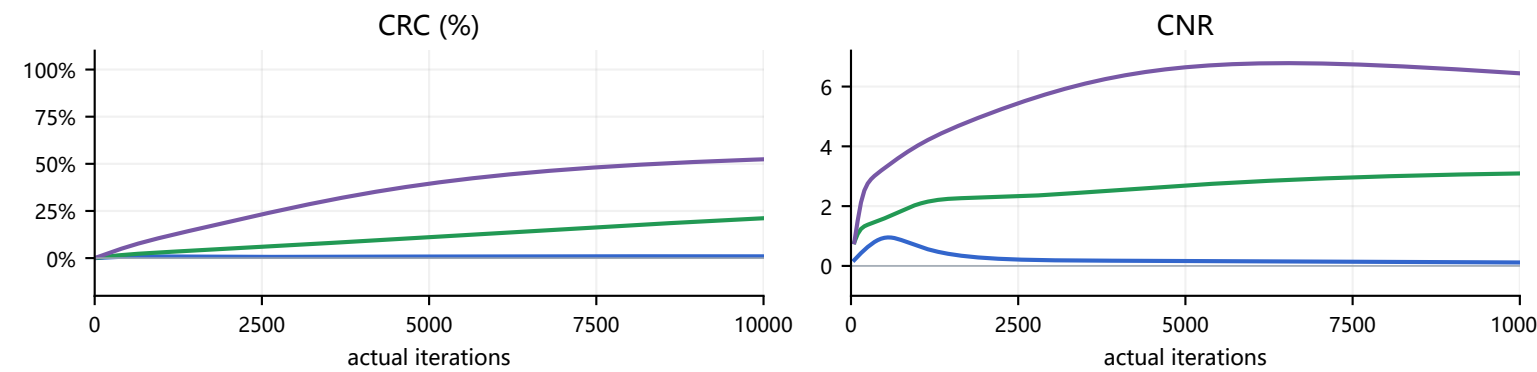
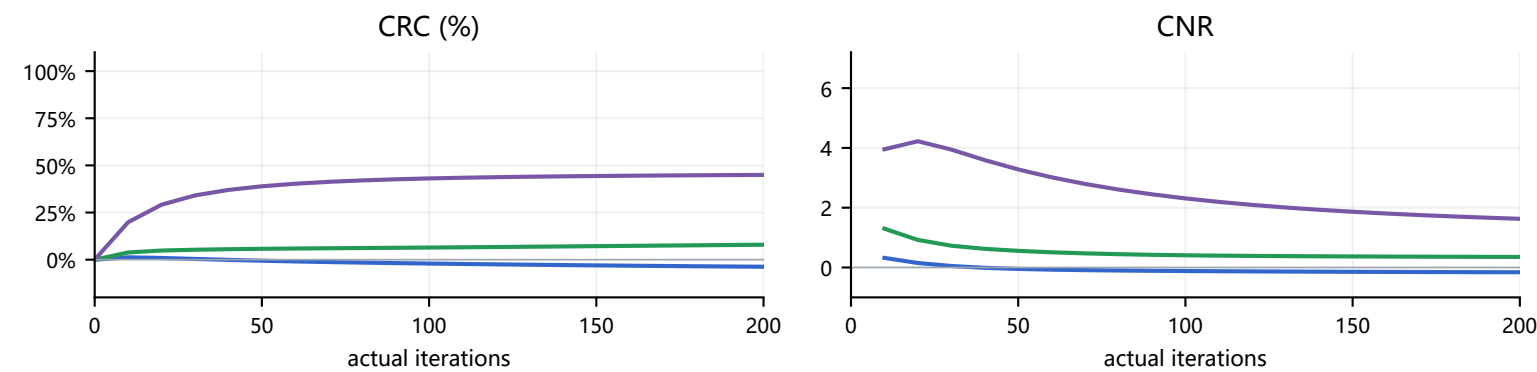
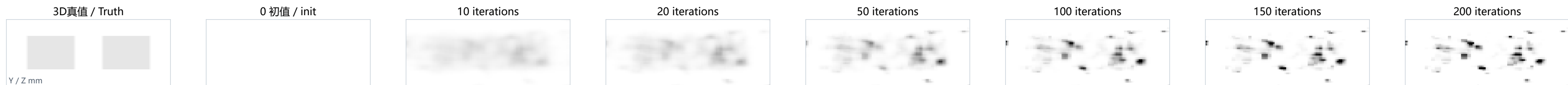
真实3D球ROI / Sphere diameter
10 mm 13 mm 17 mm 22 mm 28 mm 37 mm

整体迭代对比 / Coronal y=-1.5 mm

EHE 0-200 | JSCC 0-10000 | 9 result routes | independent iteration ranges; columns do not imply equal convergence



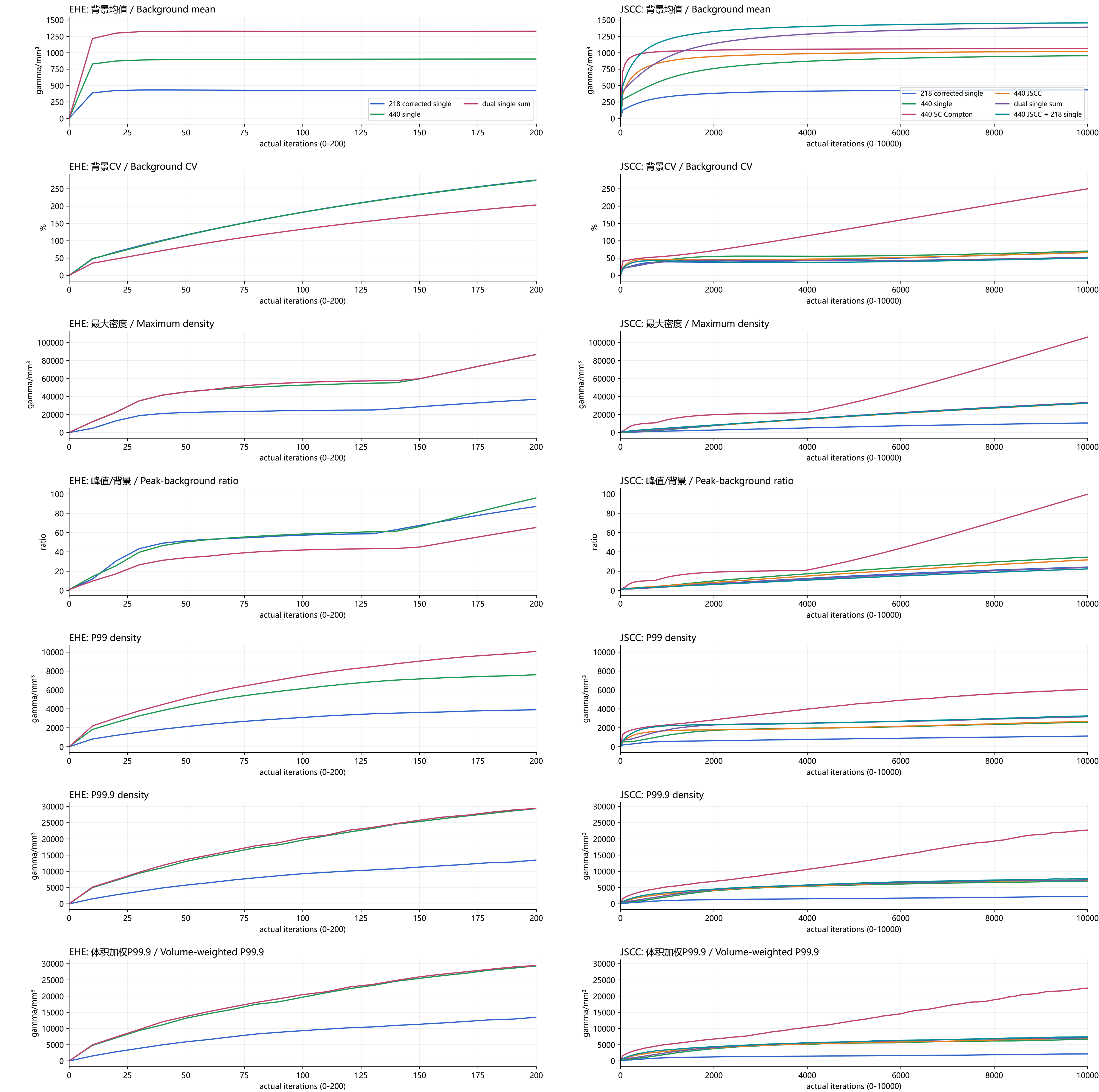
0 = known all-ones initialization (single 1, sum 2 gamma/mm³), not a saved frame; CNR at 0 undefined.
Crop 0; no smoothing or fitted gain; fixed gray: r 0-10. Metrics: native 120 mm / real 3D ROI. EHE physical audit remains HOLD.
218 background budget: EHE440 final200; JSCC440 final10000. Detector/material/count differences are not solely algorithm effects.



0 = known all-ones initialization (single 1, sum 2 gamma/mm³), not a saved frame; CNR at 0 undefined.
Crop 0; no smoothing or fitted gain; fixed gain_r 0-10. Metrics: native 120 mm / real 3D ROI. EHE physical audit remains HOLD.
218 background budget: EHE440 final200; JSCC440 final10000. Detector/material/count differences are not solely algorithm effects.

完整原生密度与噪声轨迹 / Native density and noise trajectories

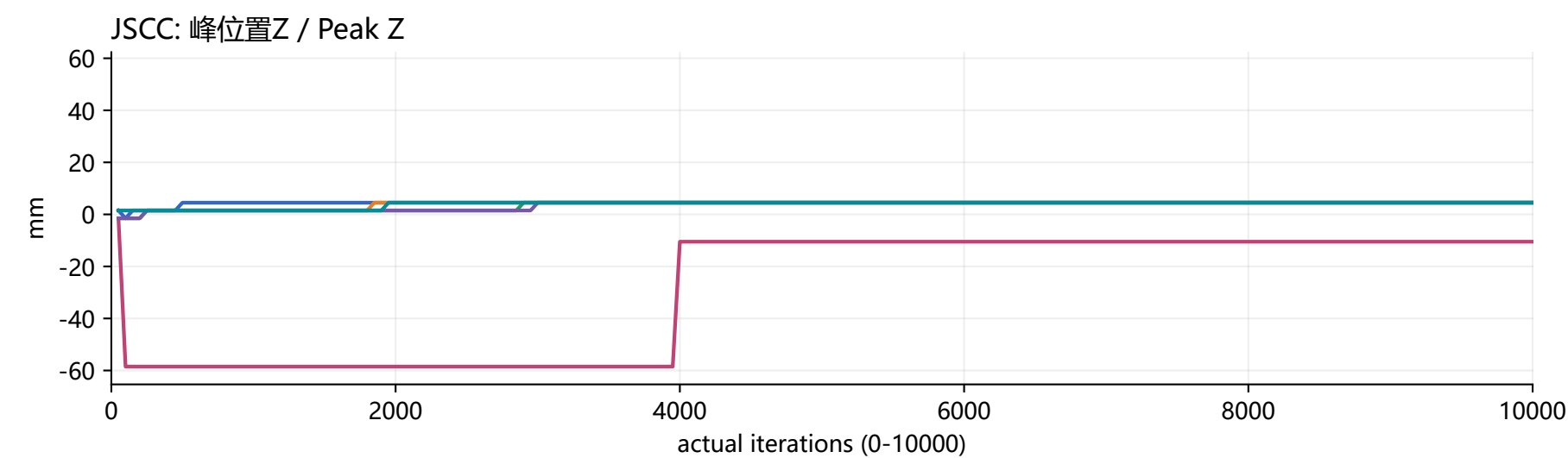
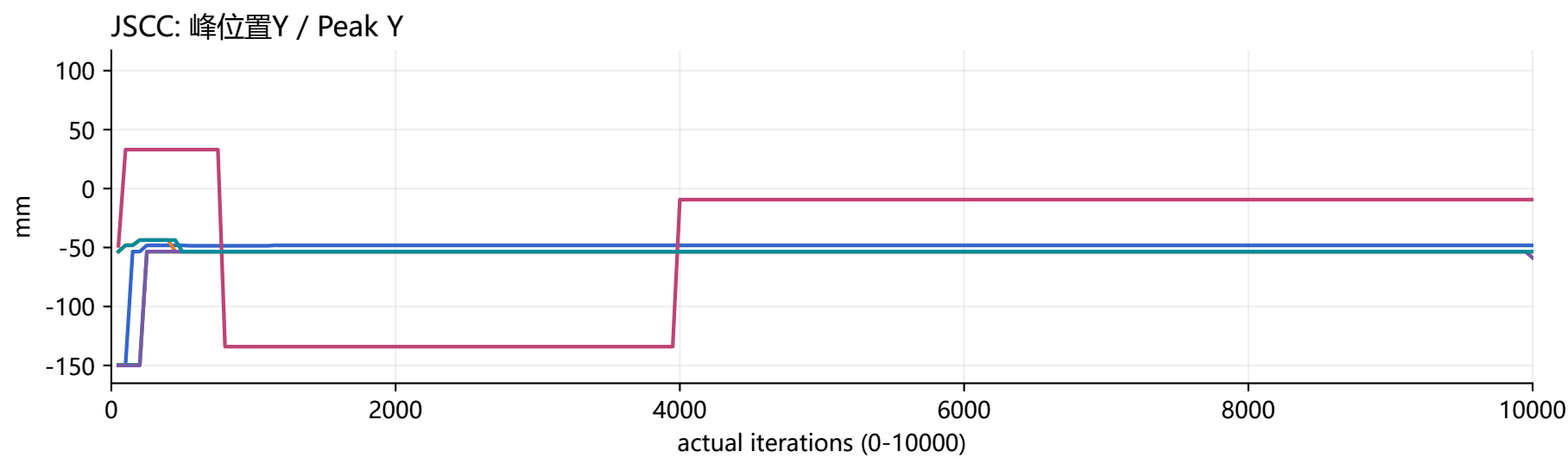
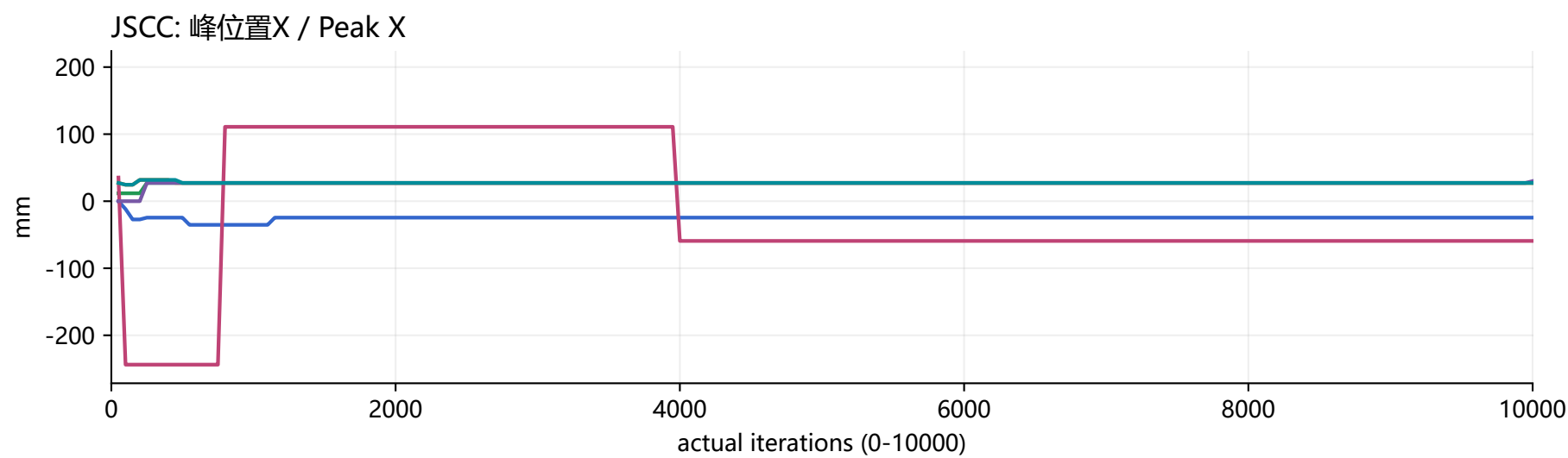
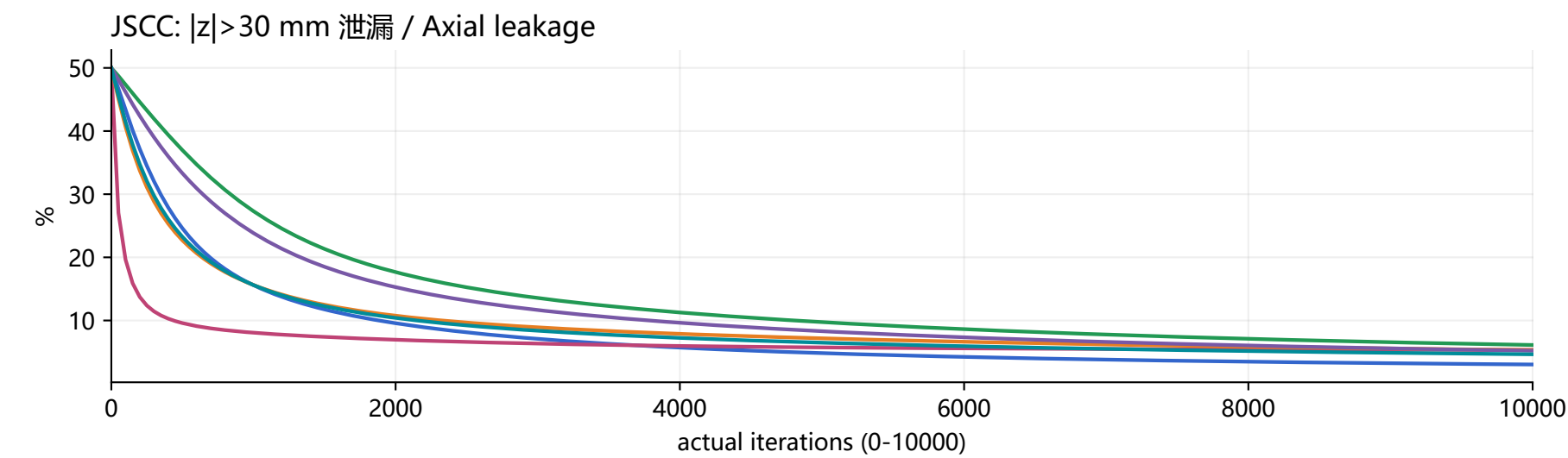
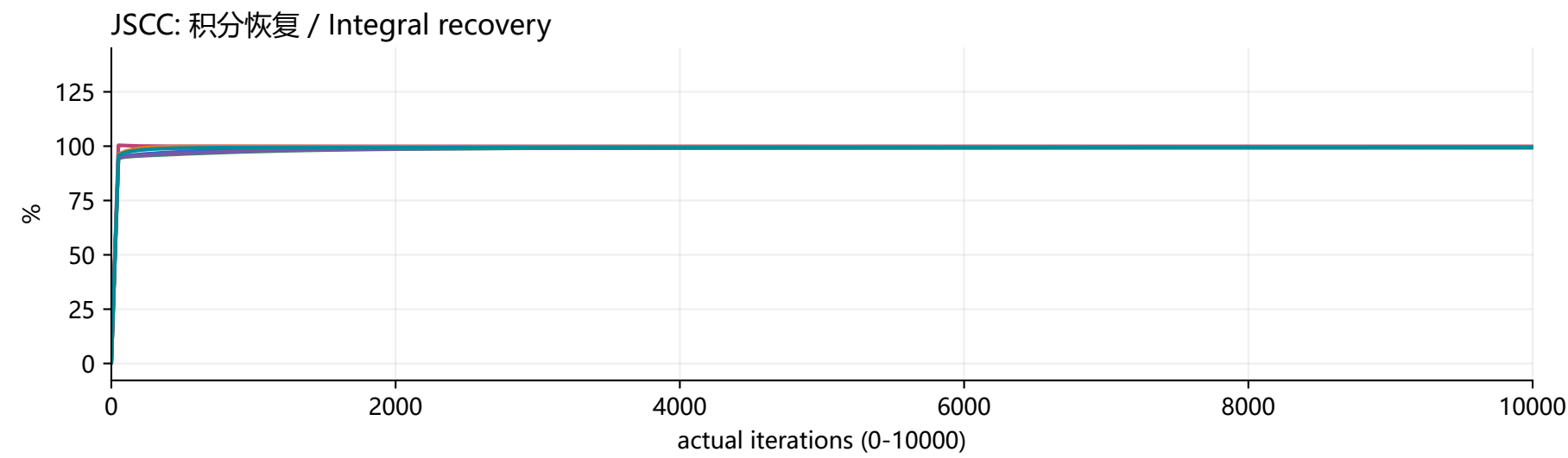
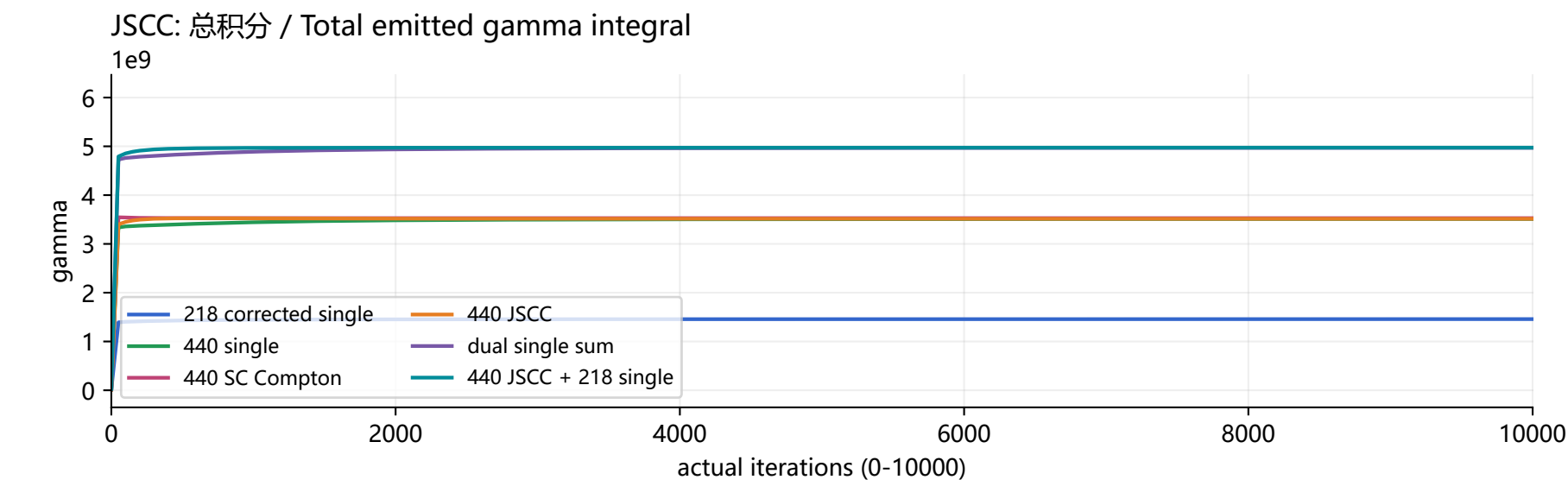
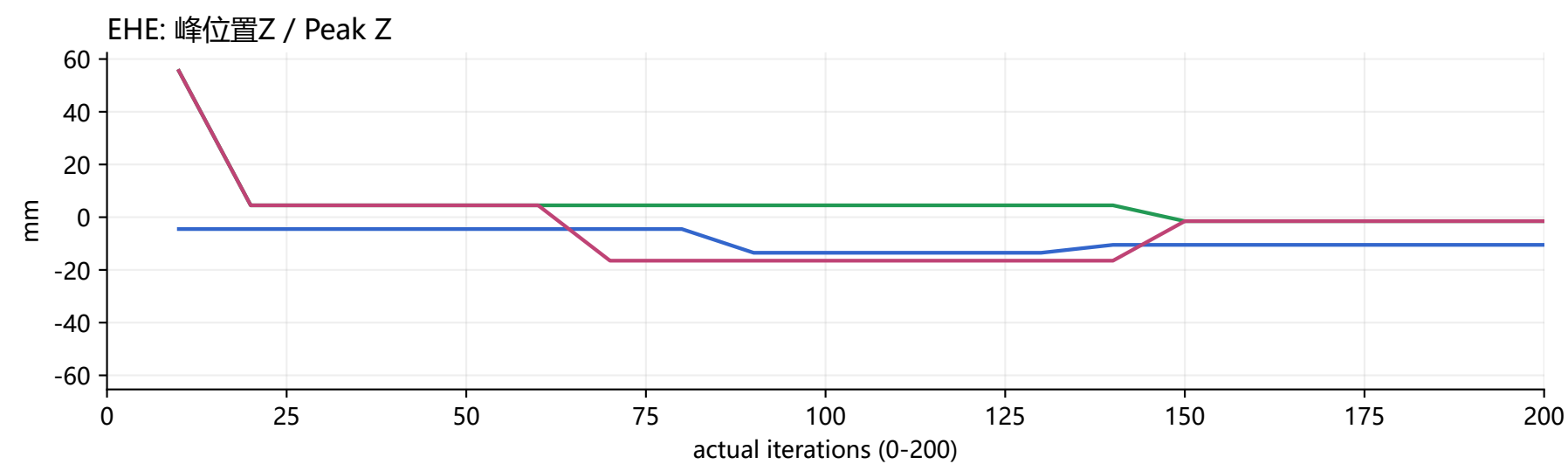
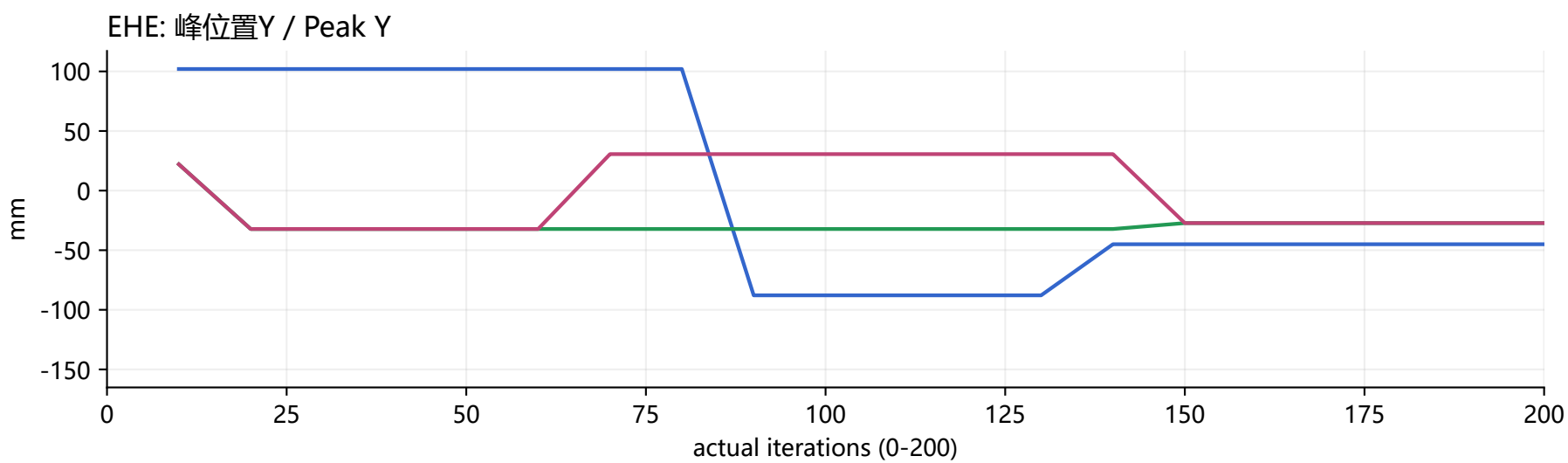
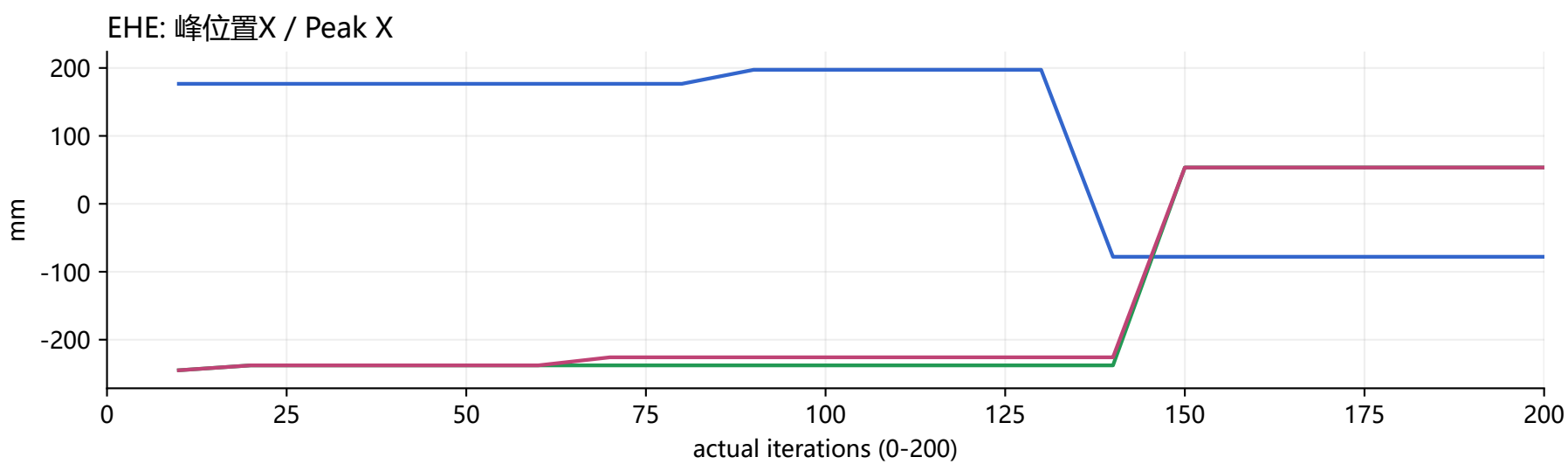
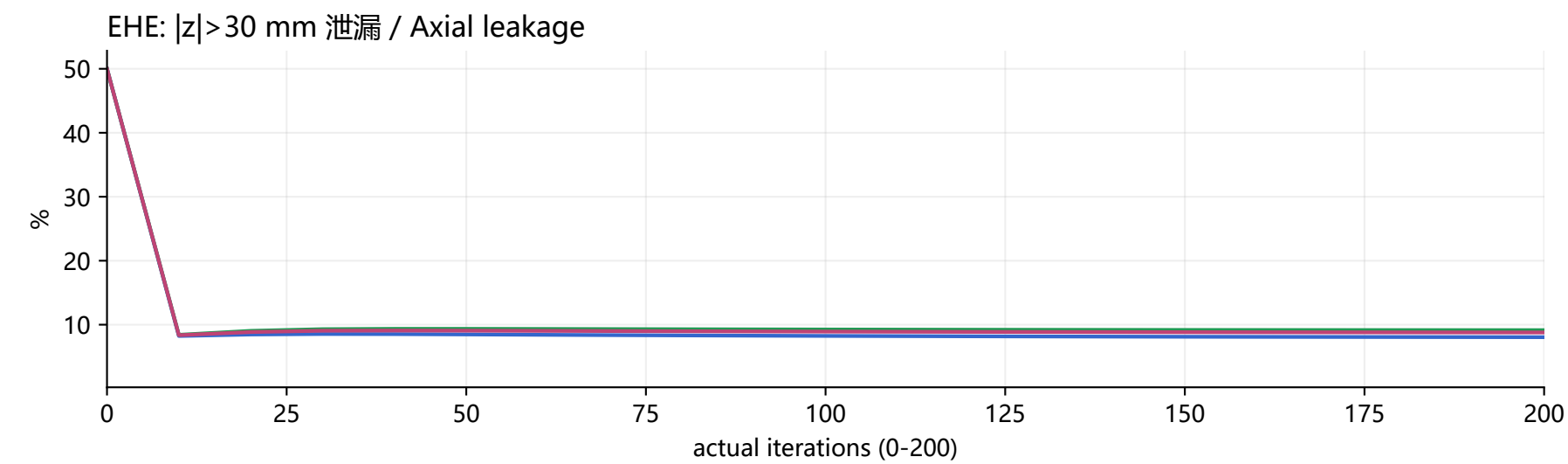
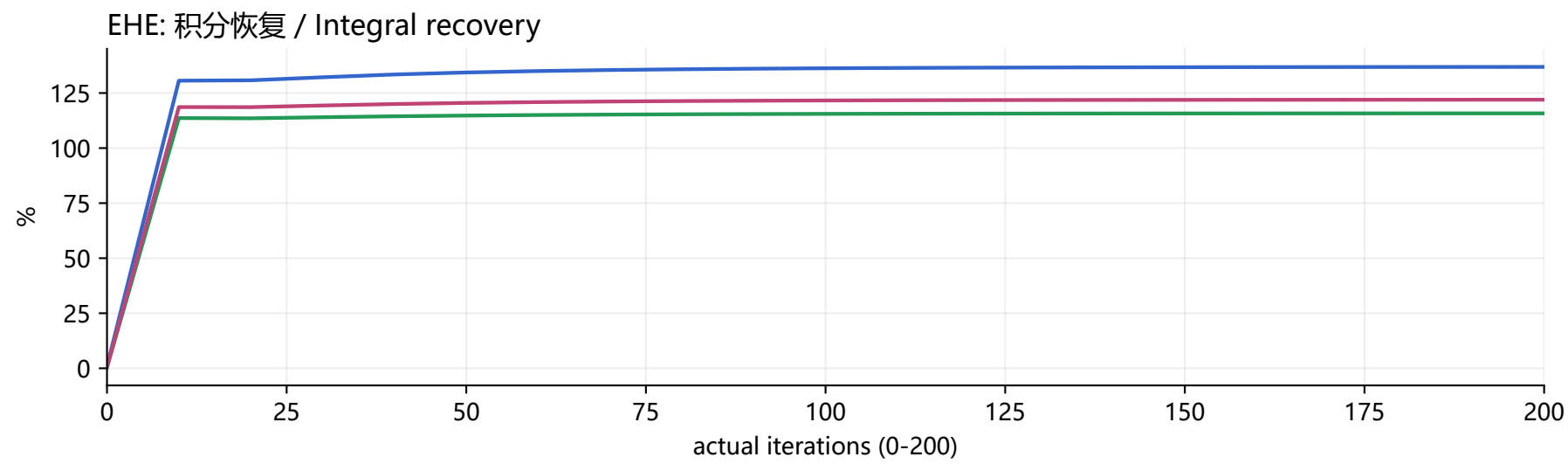
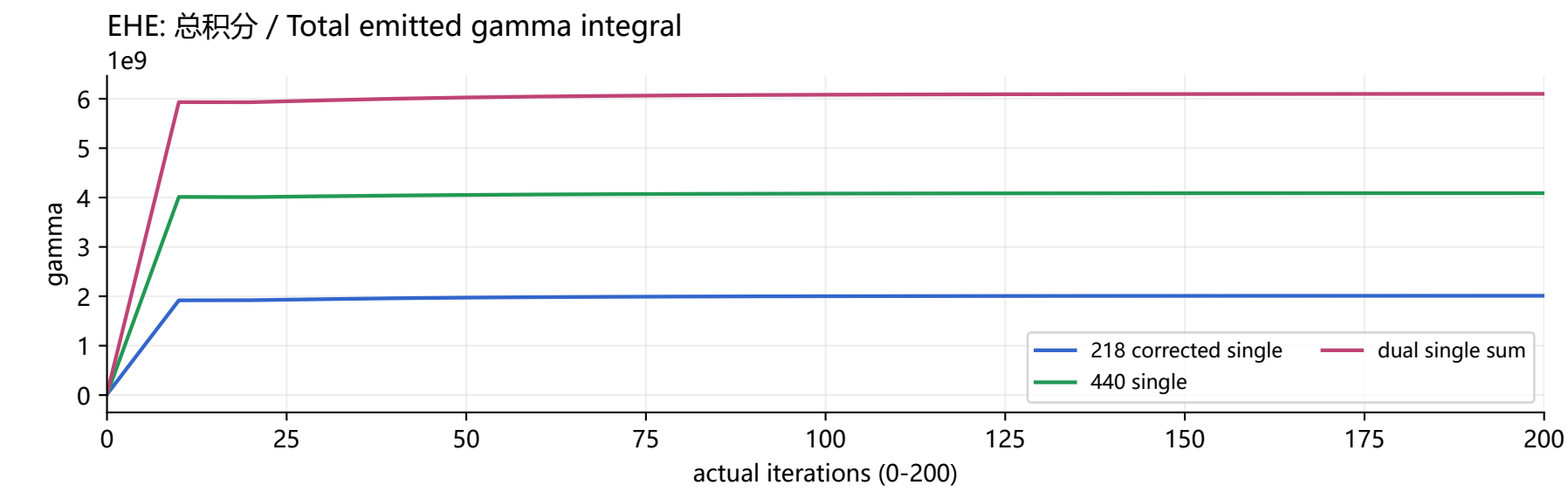
Left: EHE 0-200 (20 saved frames) | Right: JSCC 0-10000 (200 saved frames) | same vertical scale for each metric



All positive-iteration metrics copied unchanged from accepted CSVs; native complete 120 mm active cells.
0 = known uniform initialization; nonunique peak X/Y/Z omitted. No claim of equal convergence or convergence by the budget endpoint.

完整积分、泄漏与峰位置 / Integral, leakage and peak-position trajectories

Left: EHE 0-200 (20 saved frames) | Right: JSCC 0-10000 (200 saved frames) | same vertical scale for each metric



All positive-iteration metrics copied unchanged from accepted CSVs; native complete 120 mm active cells.
0 = known uniform initialization; nonunique peak X/Y/Z omitted. No claim of equal convergence or convergence by the budget endpoint.

数据与终点结果 / Data context and budget endpoints

EHE200 and JSCC10000 are separate requested budget endpoints; no equality of convergence is assumed.

同一H60双能3D真值 / Same H60 dual-energy 3D source; both 5e9 emitted gamma

System	Emitted gamma	218 window	440 window	440 to 218 measured counts
EHE	5,000,000,000	22,929	12,171	9,869 (measured primary label)
JSCC	5,000,000,000	12,314,473	5,363,190	unavailable (legacy evidence)

各自迭代预算末帧 / Final saved frame of each budget (same order of sphere, CRC and CNR)

System / iter	Result type	Sphere mm	CRC %	CNR	BG CV %	Integral %	z >30 %
EHE 200	218 corrected single	10 / 17 / 28	-3.8 / 7.9 / 45.0	-0.16 / 0.35 / 1.63	275.6	136.9	8.0
JSCC 10000	218 corrected single	10 / 17 / 28	0.9 / 21.1 / 52.4	0.11 / 3.09 / 6.44	66.9	99.3	3.0
EHE 200	440 single	13 / 22 / 37	3.4 / 59.7 / 56.2	0.15 / 2.24 / 2.27	274.4	115.8	9.1
JSCC 10000	440 single	13 / 22 / 37	18.3 / 31.8 / 38.8	3.21 / 5.51 / 5.32	70.2	99.4	6.1
JSCC 10000	440 SC Compton	13 / 22 / 37	-8.9 / 100.2 / 41.2	-0.24 / 2.32 / 1.52	250.0	99.9	5.3
JSCC 10000	440 JSCC	13 / 22 / 37	15.0 / 40.6 / 41.2	2.69 / 6.40 / 5.42	65.8	99.5	5.3
EHE 200	dual single sum	10 / 13 / 17 / 22 / 28 / 37	11.2 / -0.3 / -8.4 / 64.4 / 67.5 / 58.2	0.10 / -0.01 / -0.11 / 2.09 / 0.77 / 2.09	203.4	122.0	8.8
JSCC 10000	dual single sum	10 / 13 / 17 / 22 / 28 / 37	-14.4 / 18.2 / 32.1 / 36.6 / 78.2 / 45.7	-0.63 / 2.79 / 1.93 / 5.33 / 3.32 / 5.24	52.3	99.4	5.2
JSCC 10000	440 JSCC + 218 single	10 / 13 / 17 / 22 / 28 / 37	-14.2 / 14.7 / 33.3 / 45.9 / 74.6 / 48.3	-0.62 / 2.35 / 1.98 / 6.30 / 3.03 / 5.36	50.0	99.5	4.6

Actual counts are not matched. JSCC legacy primary-tagged cross-window counts are unavailable.
Fixed 218 additive background: EHE440200 / JSCC44010000. Original EHE physical audit remains HOLD; original method was user-authorized.